MDPI

Article

The Accessibility Trade-off: How LLM Simplification for Students with Learning Difficulties Introduces Dangerous Pedagogical Analogies

Marijela Miličević [1], Sandi Baressi Šegota [1], Ivan Lorencin [1], Darko Etinger [1]

[1] Faculty of Informatics, Juraj Dobrila University of Pula, Alda Negrija 6, 52100 Pula, Croatia; marijela.milicevic@unipu.hr (M.M.), sandi.baressi.segota@unipu.hr (S.B.Š.), ivan.lorencin@unipu.hr (I.L.), darko.etinger@unipu.hr (D.E.);

* Correspondence: ivan.lorencin@unipu.hr

Abstract

Large language models (LLMs) are increasingly used as assistive technology by students with reading and attention difficulties (e.g., dyslexia, ADHD), who rely on them as patient, private tutors capable of simplifying complex texts. It remains unclear whether this simplification compromises conceptual accuracy. This paper investigates the accessibility trade-off: the hypothesis that LLMs' attempts to be inclusive and simple shift the risk profile of their explanations, replacing technically precise statements with concrete analogies that are memorable but conceptually misleading. We introduce AccessTrap-CS, a benchmark corpus of 735 responses generated by seven open-weight LLMs for 35 core computer science concepts across five subfields. Each concept was prompted under three conditions: a standard explanation, a simplified explanation for a student with reading difficulties, and a highly simplified explanation for a student with attention difficulties. An LLM judge scored 720 responses for strategic misconception rate (SMR), dangerous analogy rate (DAR), oversimplification error rate (OER), and accuracy; a stratified subsample of 90 responses was independently annotated by two domain experts. Simplification did not significantly change the judged SMR (12.5%, 17.1%, 13.8%; $p = 0.34$). The unconditional DAR rose from 0.8% in the standard condition to 20.4% and 24.6% in the simplified conditions, but a manual coding of analogy presence on the 90-response validation subsample showed that the standard condition contained no analogies at all, while both simplified conditions contained analogies in essentially every response. The conditional DAR, computed only over responses containing an analogy, was 16.7% (B) and 10.0% (C), confirming that the unconditional DAR increase is driven mainly by analogy frequency rather than analogy quality. A generalised linear mixed model with crossed random effects for concept and model confirms the GEE results (DAR odds ratios of 36.03 and 47.34 for B vs. A and C vs. A, both $p < 0.001$; SMR odds ratios of 1.50 and 1.13, both non-significant). The human reviewers and the judge disagreed on both the magnitude and the shape of the condition effect, indicating that the absolute rates reported here are underestimates. The accessibility trade-off thus manifests chiefly as a high-rate production of analogies under accessibility prompts, a substantial fraction of which are dangerous. We propose three accessibility-first design principles for AI tutoring interfaces that signal model uncertainty and analogy status visually.

Keywords: large language models; inclusive education; strategic misconceptions; accessibility; dyslexia; ADHD; assistive technology; AI safety; computer science education; oversimplification errors; dangerous analogies

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Journal Not Specified* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Large language models (LLMs) are rapidly being adopted as assistive technology by students with reading and attention difficulties [1–3]. For learners with dyslexia, who struggle with text decoding, and students with ADHD, who face sustained attention challenges, LLMs offer a compelling value proposition: a patient, private tutor, available on demand, capable of simplifying complex academic texts in infinitely flexible ways [4,5].

Unlike human tutors, LLMs are available around the clock, never judge repeated questions, and can rephrase explanations until something clicks. It is therefore unsurprising that general-purpose assistants such as ChatGPT and Claude are increasingly used in this capacity, often without institutional oversight or teacher involvement. The same pattern is visible in computing education at large, where generative AI has moved from a curiosity to a routine study aid within a few years [6–8]. Early evidence on AI tutoring is encouraging [9,10], but it also carries a warning: unrestricted access to a generative assistant can raise performance while the assistant is available and lower it once the assistance is removed [11]. What the assistant says, and how it says it, matters for what the student actually learns.

However, this promising picture conceals a potentially dangerous blind spot. The very features that make LLMs attractive for students with learning difficulties, such as simplified language, short sentences, concrete analogies, and the avoidance of technical jargon, may inadvertently shift the risk profile of the generated explanations. We term this phenomenon the accessibility trade-off: the hypothesis that LLMs' well-intentioned attempts to be inclusive and accessible do not necessarily increase the total number of conceptual errors, but replace technically precise statements with memorable and intuitive analogies that are systematically misleading about the properties that define the target concept.

We define a strategic misconception as an explanation that is conceptually coherent, internally consistent, and often intuitively appealing, but that nonetheless contains a fundamental misunderstanding of the target concept. Unlike random errors or obvious falsehoods, strategic misconceptions are dangerous precisely because of their plausibility: a student who encounters one is not alerted to an error but is instead led to construct a mental model that is systematically wrong. This definition draws on the conceptual-change tradition in science and mathematics education [12–14] and on the misconception literature in computing [15,16], which consistently shows that intuitive but incorrect prior conceptions are among the hardest to dislodge through subsequent instruction. Two further findings make plausible-but-wrong explanations especially hazardous: fluent, easy-to-process statements are judged more likely to be true [17,18], and people routinely overestimate how well they understand mechanisms after reading a satisfying but shallow explanation [19]. For students with dyslexia and ADHD, who already expend significant cognitive effort on text decoding and attention regulation [20–22], the spare capacity for parallel fact-verification is plausibly reduced. We stress that this is a situational vulnerability arising from processing load and missing scaffolding, not a deficit in intelligence, and that it applies in some measure to any novice reading under high load.

The computer science (CS) education context makes this problem particularly acute. CS concepts are often abstract, hierarchical, and resistant to faithful simplification through everyday analogy [16,23,24]. A concept such as a semaphore, a deadlock, or virtual memory cannot be accurately captured by a single intuitive metaphor without loss of essential technical content. Yet, under pressure to simplify, LLMs frequently resort to analogies that are intuitively appealing but technically misleading. For example, comparing a semaphore to a traffic light fails to convey atomicity and counting semantics, and describing a deadlock

as two people stuck in a doorway omits resource holding and circular wait conditions. These errors are not isolated or accidental; they are systematic products of the simplification process itself. We term this subclass oversimplification errors (OER): errors that would not appear in a standard explanation but emerge exclusively as a consequence of accessibility-driven simplification.

This trade-off between warmth and accuracy is not unique to accessibility contexts. Ibrahim et al. [25] demonstrated that fine-tuning LLMs to produce warmer, more empathetic responses consistently increases error rates across model architectures, with warm models showing substantially higher rates of factual inaccuracies and sycophantic validation of incorrect user beliefs, particularly when users expressed emotional vulnerability. This fits a broader line of evidence that preference-based alignment can reward agreeable, confident-sounding answers over correct ones [26–28]. Accessibility-driven simplification represents an analogous pressure: models instructed to be maximally clear and engaging may sacrifice technical precision in ways that standard evaluation benchmarks fail to detect [29].

Prior work has documented the benefits of LLMs for students with disabilities [1,30], has studied factual errors introduced by automatic text simplification in general-purpose text [31,32], and has separately studied misconceptions in CS education [15,23]. However, to the best of our knowledge, no study has empirically investigated whether accessibility-driven simplification systematically increases the rate of strategic misconceptions in LLM-generated CS explanations, nor has any study separated the frequency of analogies from their quality under accessibility prompts. This gap is critical: if simplification primarily amplifies the frequency of dangerous analogies rather than their per-analogy danger, then current assistive technology practices may be inadvertently exposing the very population they aim to serve to a high-rate stream of memorable but conceptually misleading explanations.

Research questions and hypotheses

The study is organised around five research questions, each paired with a directional hypothesis.

- RQ1. Does accessibility-driven simplification increase the strategic misconception rate? H1: SMR(B) $>$ SMR(A) and SMR(C) $>$ SMR(A).
- RQ2. Does it increase the use of dangerous analogies? H2: DAR(C) $\geq$ DAR(B) $>$ DAR(A).
- RQ3. Is the DAR increase driven by the frequency of analogies, by their per-analogy danger, or both? We test H3a: analogy presence is higher in B and C than in A, and H3b: conditional DAR (analogies that are dangerous, computed only over responses containing an analogy) remains elevated in B and C.
- RQ4. Does simplification create errors that are absent from the standard explanation of the same concept, and do these grow with simplification intensity? H4: OER(B) $>$ 0, OER(C) $>$ 0, and OER(C) $>$ OER(B).
- RQ5. Are these effects moderated by model and CS subfield? This question is exploratory; we make no directional prediction.

In this paper, we present AccessTrap-CS, a benchmark corpus of 735 responses generated by seven open-weight LLMs across 35 core computer science concepts spanning five subfields: algorithms and data structures, programming languages and paradigms, databases, computer networks, and operating systems (7 per subfield). Each concept was prompted under three conditions: (a) a standard explanation suitable for a first-year CS student (baseline), (b) a simplified explanation simulating interaction with a student with reading difficulties (dyslexia condition), and (c) a highly simplified explanation simulating

interaction with a student with severe attention difficulties (ADHD condition). Responses were evaluated using an LLM-as-judge approach [33], with Qwen3.6:35b serving as the expert evaluator (720 of the 735 responses could be judged), and validated against a stratified human-annotated subsample of 90 responses that was also manually coded for analogy presence.

Our contributions are as follows. First, we provide empirical evidence on the accessibility trade-off, showing that the increase in dangerous analogies under simplification is driven primarily by the near-universal appearance of analogies in simplified responses rather than by a change in the per-analogy danger. Second, we provide the first conditional dangerous analogy rate for this setting, which reveals that, when an analogy is produced, it is dangerous in 10–17% of cases. Third, we introduce a novel error taxonomy: oversimplification errors (OER) as a distinct error category that emerges exclusively as a product of the simplification process. Fourth, we validate the LLM judge against two domain-expert human annotators and report both inter-annotator agreement and judge–human agreement, showing that the judge substantially underestimates error rates and disagrees with the humans about the shape of the condition effect on SMR. Fifth, we fit a generalised linear mixed model with crossed random effects for concept and model, which confirms the GEE results and shows that the DAR effect is robust to the choice of model for the repeated-measures structure. Sixth, we propose three accessibility-first design principles for AI tutoring interfaces that visually signal model uncertainty and analogy status without relying on text-based disclaimers.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 describes our methodology. Section 4 presents quantitative and qualitative results. Section 5 discusses the implications of our findings. Section 6 proposes three design principles. Section 7 acknowledges limitations. Section 8 concludes.

2. Background and Related Work

The accessibility trade-off sits at the intersection of several research streams that have developed largely independently: (1) LLMs as assistive technology for students with learning difficulties, (2) the reliability of LLMs and the pressures that trade accuracy for user-facing qualities, (3) text simplification and its pedagogical trade-offs, (4) the cognitive mechanisms that make fluent explanations persuasive and hard to verify, (5) analogies and misconceptions in computer science education, and (6) the methodology of LLM-based evaluation. This section reviews each stream and states precisely what it does and does not tell us about the accessibility trade-off.

2.1. LLMs as Assistive Technology for Dyslexia and ADHD

Dyslexia is a neurodevelopmental difference characterised by difficulties with accurate and fluent word recognition and spelling, typically associated with weaknesses in the phonological component of language [20,34]. It persists into adulthood and into higher education, where reading loads are heavy and time-constrained. ADHD is characterised by persistent inattention and/or hyperactivity-impulsivity [21,35], and meta-analyses report working-memory impairments in affected children [22]; comparable working-memory weaknesses are reported for children with learning difficulties more broadly [36]. These are group-level tendencies. Individual profiles vary widely, and the two students in our prompts are operationalisations of an interaction style (simplified language, brevity), not clinical simulations. We return to this point in Section 7.

For learners with dyslexia, LLMs can rephrase dense academic prose into shorter sentences, replace rare vocabulary with common alternatives, and maintain semantic coherence without the social cost of asking a human tutor for repeated clarification [1]. Earlier

work on text presentation and simplification for dyslexia showed that both vocabulary choices and typography measurably affect readability and understandability [4,37]. For students with ADHD, whose attention and working-memory limitations make extended technical reading challenging [21], LLMs offer on-demand summarisation and task segmentation [2]. Empirical evidence for the ADHD case specifically remains thin and largely anecdotal or self-reported; disability-centred studies of LLMs emphasise both real benefits and concerns about reliability and representation [30].

Broad overviews frame LLMs as an opportunity for personalised support alongside risks of inaccuracy and over-reliance [3]. In computing education, LLMs solve introductory programming tasks [6], and working groups document how quickly these tools have entered practice and how much of the pedagogical evidence is still missing [7,8]. Controlled studies suggest that LLM-generated help can match human-authored help on mathematics practice [9] and that a carefully designed AI tutor can outperform in-class active learning in a physics course [10]. Crucially, a field experiment in high-school mathematics found that free-form generative assistance improved performance during practice but harmed later unassisted performance, whereas a version with pedagogical safeguards largely mitigated the harm [11]. The design of what the assistant produces is therefore a first-order determinant of learning, not a cosmetic choice.

Studies on assistive text simplification have evaluated outcomes mainly through readability measures and user satisfaction [4,5], not through the conceptual accuracy of simplified technical content. No study to date has examined whether LLM-generated simplifications for learners with reading or attention difficulties introduce novel errors that are absent in standard explanations, nor has any study separated the frequency of analogies from their per-analogy danger. The gap is non-trivial: the mechanisms that make simplification beneficial (reduced decoding effort, shorter sentences, concrete analogies) may simultaneously suppress the nuanced qualifications and precise technical language that prevent conceptual misunderstanding.

2.2. LLM Reliability, Sycophancy, and the Warmth–Accuracy Trade-Off

LLMs produce fluent text that is not necessarily faithful to fact or to a source [38,39]. Two properties of this literature matter here. First, errors are not confined to obscure facts: models can reproduce popular misconceptions, and on the TruthfulQA benchmark the largest models were often among the least truthful because they imitate widespread human falsehoods more faithfully [29]. This is directly relevant to analogies, which are precisely the popular, widely repeated framings of technical ideas. Second, factual quality is prompt-dependent: the same model can be accurate under one instruction and misleading under another.

Instruction-following and preference-based training optimise for outputs that human raters approve [26]. Approval and correctness are correlated but not identical. Models trained this way exhibit sycophancy, tailoring answers to the beliefs or expectations of the user, and preference data itself can reward it [27,28].

Ibrahim et al. [25] demonstrated that fine-tuning models to produce warmer, more empathetic responses increases error rates across architectural families, with warm models showing a statistically significant tendency toward sycophantic validation of incorrect user beliefs. This trade-off is not an implementation flaw but a plausible consequence of alignment pressure: models optimised for human preference judgments learn that users respond positively to agreeable, confident-sounding explanations, even when those explanations are technically incomplete.

Where warmth optimisation prioritises affective alignment, simplification prioritises cognitive accessibility. The two are not identical: a simplified explanation need not be warm,

and a warm explanation need not be simple. However, they share a common mechanism: both pressures push models away from precise, qualified, context-dependent technical statements and toward concrete, declarative, analogy-laden statements that are easier to process but more likely to be wrong in subtle ways.

We propose the term oversimplification error (OER) to capture a specific subclass of simplification-induced inaccuracies: errors that would not appear in a standard, technically precise explanation of the same concept but emerge exclusively as a product of the simplification process. OERs are distinct from random factual errors, such as misstating the year of an algorithm's invention, in that they arise systematically from the interaction between the concept's inherent complexity and the pressure to reduce that complexity. A standard explanation of semaphores, for example, would specify atomicity, counting semantics, and the distinction between blocking and busy-waiting. A simplified explanation that omits atomicity is not merely incomplete; it actively misleads by suggesting that semaphores are just integer variables with special names.

A second distinction, which we introduce in this work, separates the frequency of analogies from their danger. A prompt that explicitly requests analogies will trivially increase the unconditional dangerous analogy rate, even if the model's per-analogy accuracy is unchanged. The conditional dangerous analogy rate, computed only over responses that contain at least one analogy, is the appropriate measure of per-analogy quality. We report both rates in Section 4.

Widely used evaluations measure knowledge or truthfulness under a fixed, neutral prompt [29]. A model can therefore score well on standard tests and still degrade when told to speak to a particular kind of learner. AccessTrap-CS is designed to expose exactly this audience-conditioned degradation.

A natural response is to have models flag their own uncertainty. Verbalised model confidence is often overconfident, and better estimators require additional computation such as semantic entropy or consistency across samples. This matters for the interface proposals in Section 6: a visual uncertainty signal is only as trustworthy as the estimator behind it.

2.3. Text Simplification: Pedagogical Benefits and Technical Risks

Automatic text simplification has a long history in natural language processing, evolving from rule-based systems applying syntactic transformations and lexical substitution to neural sequence-to-sequence approaches and, more recently, LLMs [5]. Evaluation has traditionally focused on three dimensions: readability (measured by standard formulas such as Flesch Reading Ease [40] and Flesch-Kincaid grade level), semantic preservation (measured by entailment or meaning similarity), and grammaticality or fluency.

Simplification is not information-preserving by default. Work on factuality in simplification shows that models frequently insert errors or drop information, and that the errors are of kinds standard metrics do not catch [31]; Trienes et al. [32] characterise information loss in simplification explicitly and show that it can be surfaced through question-answering. These studies concern general-purpose text (news, encyclopedic and medical content). Our contribution is to move to technical CS concepts, where the information that gets lost tends to be exactly the boundary condition that defines the concept.

Conventional simplification evaluation assumes a monotonic relationship between simplification and information retention: simpler texts should preserve the truth conditions of the original while reducing linguistic complexity. For many domains, including news articles, consumer health information, and government documents, this assumption holds reasonably well. For computer science education, it breaks down. CS concepts are not facts about the world but abstract, relational structures with precise definitions and boundary

conditions. A linked list is not merely a chain of connected items, as that description also fits a rope, a queue, or a biological pathway. The pedagogical challenge is to convey both the intuitive metaphor and the precise conditions under which the metaphor fails. Good CS instruction does not replace technical precision with analogy; it layers analogy on top of precision, using metaphor as a scaffold, not a substitute.

Even in the human-authored text literature, easier does not equal better for learning. Coherence-enhancing text revisions helped low-knowledge readers on shallow tests but limited deeper learning for readers with more knowledge [41], and the expertise reversal effect shows that support that helps novices can become a hindrance as knowledge grows [42]. Some difficulty is desirable, because effortful processing supports durable understanding [43]. The implication is not that learners with reading or attention difficulties should face inaccessible text, but that accessibility should reduce extraneous load (decoding, layout, jargon) while leaving intact the germane content (conditions, invariants, exceptions) [44].

LLMs asked to simplify lack this pedagogical awareness. Under pressure to produce simpler output, they can omit boundary conditions, elide exceptions, and promote analogies from the intuitive level to the definitional level. The result is not a simplified version of the same concept but a different, incorrect concept that happens to share surface features with the target. A student who learns that a deadlock is two people stuck in a doorway has not learned a simplified but essentially correct version of deadlock; they have learned a misconception that will later need to be unlearned. Crucially, the frequency of this failure mode and the per-analogy danger of the analogy are distinct quantities, and a study that does not separate them cannot tell whether the model produces more analogies or worse analogies.

2.4. Why Fluent Explanations Mislead: Cognitive Foundations of Vulnerability

The claim that simplified explanations are especially hazardous for learners with reading or attention difficulties rests on four well-established cognitive mechanisms. We present them here as a theoretical motivation; our experiment does not measure them.

Working memory is severely limited [45], and cognitive load theory distinguishes intrinsic, extraneous, and germane load [44]. Automaticity theory holds that when word recognition is not automatic, decoding competes with comprehension for the same limited resources [46]. For a reader with dyslexia, decoding remains effortful longer; for a reader with ADHD, sustaining attention consumes regulation resources [21,22]. Both leave less spare capacity for what a critical reader does on top of comprehension: checking claims against prior knowledge, noticing what is missing, and asking whether an analogy really holds.

Statements that are easy to process are judged more likely to be true and elicit higher confidence [17,18]. A response deliberately engineered to be easy to read therefore also carries a built-in truth cue, whether or not the content deserves it. This is a second-order side effect of accessible design: the same property that reduces decoding cost also lowers the reader's guard.

People typically feel they understand causal systems far better than they can actually explain them, and the feeling is punctured only when they attempt to produce the explanation [19]. A short, vivid, self-contained explanation provides the feeling of understanding without the test, which is the condition under which an incorrect mental model can consolidate without challenge.

Dual-process accounts distinguish fast, effortless, intuitive judgement from slow, effortful checking [47]. Accepting a fluent explanation is the intuitive default; verification is the costly step. Human-automation research shows that people over-rely on automated

aids, particularly when verification is expensive, and that interface interventions which force engagement can reduce over-reliance, at the price of extra effort [48]. We revisit this trade-off in Section 6.

Finally, interesting but tangential material can impair learning of the core content [49]. Condition C, which explicitly asks for a memorable and engaging analogy, is exactly the situation in which the vividness of the analogy may compete with, rather than support, the technical content.

2.5. Analogies and Misconceptions in Computer Science Education

Learners do not arrive as blank slates: they build intuitive models, and instruction must confront and restructure them [12,14]. Misconceptions that are consistent with everyday experience are resistant to correction, because a wrong model that successfully explains a subset of observations is unlikely to be abandoned without clear counterexamples [13]. This makes the first exposure to a concept critical.

The study of misconceptions in CS education is well established. Qian and Lehman [15] provide a comprehensive review, identifying common incorrect mental models across introductory programming, such as variable assignment as mathematical equality, the countdown misconception in loops, confusion between reference and value semantics, as well as data structures, such as believing that binary search trees are inherently balanced, and concurrency, such as thinking that race conditions are deterministic or that semaphores prevent all interleavings. Foundational work documented that novice bugs are often language-independent conceptual errors [24], that many misconceptions stem from an inaccurate mental model of the machine [16], and that students' understanding of synchronisation evolves through intermediate, faulty models. These resources provide the ground truth logic for our concept selection and error coding, but they describe misconceptions held by human novices, not those induced by generated explanations.

Analogy is a central instrument of explanation. Structure-mapping theory holds that a good analogy transfers relational structure, not surface attributes [50]. A traffic-light analogy for a semaphore conveys an attribute (something signals go/stop) but not the relational structure (atomic counting operations coordinating concurrent processes). Analogies are widespread in science and computing teaching, and can improve learning of programming when they match the target model. But it has long been recognised that analogies produce analogy-induced misconceptions, because learners import features of the source that the target does not have [51]. Recommended practice requires explicitly mapping the shared features and, crucially, stating where the analogy breaks down [52]; multiple analogies with complementary strengths are proposed as an antidote to single-analogy misconceptions [51].

Condition C of our study requests one single, vivid analogy, which is exactly the situation the analogy literature warns against: a single analogy, with no stated limits, presented as the core of the explanation. Stone et al. [23] recently extended the analogy analysis to LLM-generated educational content, documenting that common analogies produced by GPT-4 for CS concepts frequently misrepresent technical properties. However, their study used standard prompting without explicit accessibility constraints, did not compare simplification conditions, and did not separate analogy frequency from analogy quality. Our work builds directly on it by systematically varying simplification pressure, measuring the resulting amplification of dangerous analogies, and separating the frequency-driven from the quality-driven components of that amplification.

2.6. LLM-as-Judge Evaluation

Scaling up expert judgement of open-ended text has led to the widespread use of strong LLMs as evaluators. Strong judges can reach agreement with human raters at a level comparable to human–human agreement on chat tasks [33], and rubric-based prompting improves alignment with humans on generation quality [53]. The approach has known biases that shape our design: sensitivity to the position of candidate answers [54], verbosity preferences, and a preference for the judge's own outputs [55]. Section 3 describes how we address these biases and how we validate the judge against human experts; Section 7 discusses the remaining risks.

2.7. The Accessibility Trade-off: A Formal Hypothesis

Synthesising these streams, we formalise the accessibility trade-off as a testable hypothesis. For a given concept c in computer science, let $E_s(c)$ be an LLM-generated explanation under a standard prompt (baseline) and $E_a(c)$ be an explanation under an accessibility prompt requesting simplification for a student with reading or attention difficulties. The strategic misconception rate $\mathrm{SMR}(E)$ is the probability that E contains a conceptually coherent but fundamentally incorrect understanding of c. The oversimplification error rate $\mathrm{OER}(E)$ is the probability that E contains an error present in E_a but absent from E_s for the same concept. The dangerous analogy rate $\mathrm{DAR}(E)$ is the unconditional probability that E contains a dangerous analogy, and the conditional dangerous analogy rate $\mathrm{DAR}_{\mathrm{cond}}(E)$ is the same probability restricted to responses that contain at least one analogy.

The accessibility trade-off hypothesis states that simplification does not uniformly increase the general strategic misconception rate, but shifts the risk profile towards analogy-driven errors:

$$\mathrm{SMR}(E_a) \gtrsim \mathrm{SMR}(E_s) \quad \text{and} \quad \mathrm{DAR}(E_a) \gg \mathrm{DAR}(E_s), \quad \text{with } \mathrm{OER}(E_a) > 0, \tag{1}$$

and it further decomposes the DAR gap into two components:

$$\underbrace{\mathrm{DAR}(E_a)}_{\text{unconditional}} = \underbrace{p_{\text{analogy}}(E_a)}_{\text{analogy frequency}} \times \underbrace{\mathrm{DAR}_{\mathrm{cond}}(E_a)}_{\text{per-analogy danger}}. \tag{2}$$

The unconditional DAR increase can therefore be driven by analogy frequency, by per-analogy danger, or by both. AccessTrap-CS measures each component separately.

This hypothesis does not imply that simplification is inherently harmful or that LLMs should not be used as assistive technology. Rather, it predicts a previously undocumented failure mode that current evaluation practices do not detect. The AccessTrap-CS benchmark described in Section 3 was designed to test this hypothesis across models, subfields, and simplification intensities.

3. Materials and Methods

This section describes the benchmark design, concept selection, model selection, prompt engineering, data generation procedure, evaluation framework, human evaluation protocol, readability metrics, and statistical analysis plan. For each design decision we state the rationale, because in a benchmark study the choices are part of the result.

3.1. Benchmark Design Overview

The AccessTrap-CS benchmark tests the accessibility trade-off hypothesis across three dimensions: model (7 LLMs), CS subfield (5 subfields), and simplification intensity (3 conditions). The design is fully crossed: each of the 35 concepts is posed to each of the 7

models under each of the 3 conditions, and each cell produces a single response, yielding 735 responses. All 735 were used in the readability analysis; 720 could be evaluated by the LLM judge (15 responses from five concept-model groups, concentrated in the reasoning models, were lost to unrecoverable parsing failures). The 35 concepts and 7 models yield 245 concept-model pairs, each observed under three conditions.

$$N_{\text{total}} = C \times M \times K = 35 \times 7 \times 3 = 735 \tag{3}$$

where $C = 35$ concepts, $M = 7$ models, and $K = 3$ conditions. Because the same concept-model pair appears in all three conditions, condition is a within-pair (repeated) factor. This matters for analysis and is what allows the oversimplification error to be defined as a contrast within a pair. Figure 1 shows the overall pipeline.

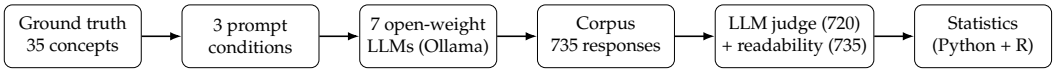


Figure 1. AccessTrap-CS pipeline: from concept ground truth to statistical analysis.

The pipeline proceeds in six stages. Ground-truth definitions for the 35 concepts are first paired with the three prompt conditions, then passed to the seven open-weight LLMs served locally through Ollama. The resulting responses are aggregated into the 735-response corpus, which is then scored by the LLM judge and analysed with the readability metrics. Finally, the scored corpus is analysed statistically in Python (descriptive tests, GEE, conditional DAR) and in R (generalised linear mixed models with crossed random effects). Each of these stages is described in detail in the subsections that follow.

3.2. Concept Selection

A total of 35 core computer science concepts were selected across five subfields: Algorithms and Data Structures, Programming Languages and Paradigms, Database Systems, Computer Networks, and Operating Systems (7 concepts per subfield). The balanced design ensures that subfield comparisons are not driven by unequal sample sizes. Concepts were drawn from standard core undergraduate curricula following the ACM/IEEE-CS/AAAI Computer Science Curricula [56]. Each concept was paired with a ground-truth definition and a reference source from standard textbooks. The ground truth serves as the judge's anchor: without it, the judge would have to rely on its own parametric knowledge, which is exactly the resource whose reliability is under test. The complete list appears in Appendix A.

3.3. Model Selection

Seven open-weight LLMs were selected to ensure reproducibility and to cover a range of model families, parameter scales, and architectural designs. Open weights allow anyone to rerun the whole pipeline locally, without dependence on a vendor's changing API. All models were accessed locally via Ollama running on the Supek HPC cluster. Table 1 summarises the selected models.

Table 1. Selected open-weight LLMs. Ollama tags and parameter counts were verified with `ollama show`.

Model (Ollama tag)	Family	Parameters
mistral:7b	Mistral AI	7B (dense)
deepseek-r1:8b	DeepSeek (distilled)	8B (dense, reasoning)
ministral-3:14b	Mistral AI	14B (dense)
phi4:14b	Microsoft	14B (dense)
gpt-oss:20b	OpenAI (open-weight)	≈21B total (MoE, reasoning)
gemma4:31b	Google	31B (dense)
deepseek-r1:32b	DeepSeek (distilled)	32B (dense, reasoning)

Models span parameter scales from 7B to 32B, include three models with explicit reasoning traces (deepseek-r1:8b, deepseek-r1:32b and gpt-oss:20b), and cover five distinct model families. Model family and scale are therefore partly confounded in this design. Ollama serves quantised weights by default; we therefore treat model behaviour as that of the quantised builds actually used, and report the exact Ollama tags, quantisation levels, and model digests in the repository. All inference was performed with temperature $T = 0.7$ and `num_predict` = 512 tokens.

The temperature $T = 0.7$ was chosen to approximate typical chat deployment, where users interact with sampled rather than greedy outputs; the price is run-to-run variability, which we discuss as a limitation in Section 7.

3.4. Prompt Engineering and Condition Design

Three prompt templates were designed corresponding to three simplification intensities. All three share the same structure (an instruction followed by `Answer in English.` and `Concept: [CONCEPT]`), so that the only variable is the instruction. The prompts operationalise two recurring accessibility recommendations: reducing decoding load (short sentences, plain vocabulary, no formal notation) and reducing sustained-attention demand (brevity, a single memorable hook).

3.4.1. Condition A: Standard Explanation (Baseline)

The baseline prompt requests an accurate and clear explanation suitable for a first-year computer science student. No simplification constraints are applied. It defines the reference against which errors introduced by simplification are measured.

```
Explain the following computer science concept accurately and clearly,
suitable for a first-year computer science student.

Answer in English.

Concept: [CONCEPT]
```

3.4.2. Condition B: Simplified Explanation (Reading Difficulties)

The simplified prompt instructs the model to address a student with reading difficulties. Constraints include simplified language, short sentences, concrete analogies, avoidance of code and formal definitions, and minimal technical jargon. This condition targets the decoding-load mechanism.

```
Explain the following computer science concept to a student who has
reading difficulties. Use simplified language, short sentences, and
concrete analogies instead of code or formal definitions. Avoid
technical jargon where possible.
```

```
Answer in English.

Concept: [CONCEPT]
```

3.4.3. Condition C: Highly Simplified Explanation (Attention Difficulties)

The highly simplified prompt instructs the model to address a student with significant attention difficulties. Constraints include a maximum of five sentences, a single vivid analogy that captures the core idea, and emphasis on memorability and engagement. This condition targets the sustained-attention mechanism and is intended as the higher-intensity condition, because it removes even the space in which qualifications could be placed.

```
Explain the following computer science concept to a student with
significant attention difficulties. Keep the explanation within five
sentences. Use one single, vivid analogy that captures the core idea.
Make it memorable and engaging.

Answer in English.

Concept: [CONCEPT]
```

3.4.4. Construct Validity of the Conditions

Three properties of the prompt design need to be stated explicitly. First, the conditions are bundles of constraints: B and C differ from A, and from each other, in several respects at once (audience label, register, length, number of analogies, instruction to avoid formal definitions). The study therefore tests whether the bundle typical of accessibility-oriented simplification is risky, not the causal contribution of any single element. This matters for the interpretation of the frequency-versus-quality decomposition of the DAR gap: we can measure both components, but we cannot attribute either to a single prompt element. Second, only condition A contains the word accurately; an accuracy cue is absent in B and C, mirroring how real users write requests but confounding simplification with the absence of that cue. Third, the instruction in B to avoid formal definitions directly discourages the very precision whose absence we count as error, and the instruction in B and C to use concrete analogies directly increases the frequency of analogies. We treat this as the realistic operational form of the trade-off (users and teachers do ask for these things) but acknowledge it in Section 7. Finally, the labels dyslexia condition and ADHD condition denote the prompt persona, not the diagnostic status of any real person.

Table 2 summarises the constraint bundle of each condition. The table makes explicit that B and C differ from A in several respects at once and that C differs from B in the additional constraints of a hard length limit and a single-analogy requirement. This design choice is deliberate: the study tests the risk profile of the bundle that teachers and students actually request, not the causal contribution of any single prompt element.

Table 2. Design of the three prompt conditions. Entries indicate whether the constraint is present (✓), absent (×), or partially present (∼). The conditions are bundles: no single element is isolated.

Constraint	A	B	C
Accuracy cue ("accurately")	✓	×	×
Simplified language	×	✓	✓
Short sentences	×	✓	✓
Concrete analogies requested	×	✓	✓
Avoid code and formal definitions	×	✓	×
Maximum length (5 sentences)	×	×	✓
Single vivid analogy required	×	×	✓
Audience label (reading difficulties)	×	✓	×
Audience label (attention difficulties)	×	×	✓

We emphasise that this is a model-side audit study rather than a user study. The prompts operationalise an interaction style; they do not simulate the cognitive processing of a real learner, nor do they claim that simplification is the only or the best accommodation for the populations named in the prompts. Direct user studies with neurodivergent learners are essential for evaluating comprehension and learning outcomes, but model-side audits such as ours are a necessary prerequisite for ensuring system safety before exposing vulnerable learner populations to potential domain misinformation.

3.5. Data Generation Procedure

Data generation was automated using a Python script that iterates over all combinations of concepts, conditions, and models.

3.5.1. Input Format

The script reads `ground_truth.csv` (semicolon-delimited) containing the following columns: `ID`, `Domain`, `Concept`, `Ground_Truth`, `Reference`.

3.5.2. Generation Loop

For each concept, the script formats the three prompt templates by substituting the concept name. For each of the 7 models, it sends a POST request to the Ollama API at `http://localhost:11434/api/generate`, collects the response text, and appends the response to `corpus.csv` with metadata. A 1-second pause between requests prevents API overload. Failed requests are retried with a 5-second delay. No system prompt, conversation history, or few-shot examples were used, so each response reflects a single-turn interaction.

3.5.3. Output Format

The complete corpus `corpus.csv` contains 735 rows with the following schema: `concept_id`, `domain`, `concept`, `condition`, `model`, `prompt`, `response`, `ground_truth`, `reference`, `timestamp`.

3.6. Evaluation Framework

Responses were evaluated using an LLM-as-judge approach [33] with Qwen3.6:35b serving as the expert evaluator. The evaluation script groups responses by concept and model, then presents all three conditions (A, B, C) to the judge simultaneously together with the ground truth. Joint presentation is required for the OER judgement, which is by definition a comparison between a simplified response and the baseline for the same pair; the cost is that the judge's decisions on one condition may be influenced by the others (Section 7).

3.6.1. Judge Model Selection

Qwen3.6:35b was chosen as the judge for three reasons: (a) it is among the stronger open-weight models that can be served locally, (b) its open-weight license ensures reproducibility, and (c) it is architecturally and organisationally distinct from the seven generators. Because the judge is a different model from all generators, no self-evaluation bias arises from a model judging its own outputs. The judge is configured with temperature $T = 0.0$ for deterministic outputs and `num_predict` = 1024 tokens.

3.6.2. Error Taxonomy

Four evaluation dimensions are defined.

Strategic Misconception (SM). A Boolean indicator of whether the explanation contains a subtle but conceptually significant error that sounds plausible but is fundamentally wrong. The defining features are plausibility and consequence: a misstated date or a typographical slip is an error but not a strategic misconception, while claiming that a deadlock is merely processes waiting a long time is.

Dangerous Analogy (DA). A Boolean indicator of whether the explanation uses an analogy that is intuitive but misrepresents a key technical property of the concept. An analogy is not dangerous merely for being incomplete; it is dangerous if it invites an inference that is false for the target (e.g., a semaphore as a traffic light invites the inference that a semaphore is a simple binary flag with no atomicity).

Oversimplification Error (OER). A Boolean indicator of whether the explanation contains an error that would not appear in a standard explanation, i.e., an error that exists only because of the simplification process. By construction, OER is defined only for conditions B and C, since the reference is condition A.

Accuracy Score. A Likert scale from 1 to 5, where 1 = completely wrong and 5 = perfectly accurate. It complements the Boolean indicators by capturing graded degradation that does not cross the threshold of a strategic misconception.

The three Boolean indicators overlap by design (a dangerous analogy is a common route to a strategic misconception, and an oversimplification error is typically also a misconception) but are not identical, and they are analysed both separately and jointly.

Analogy presence. A separate Boolean indicator, coded manually on the human validation subsample, of whether the response contains at least one analogy, metaphor, or simile. This indicator is not part of the judge evaluation; it is defined only for the 90 responses in the validation subsample, and it is used to compute the conditional dangerous analogy rate (Section 3.9). A response was coded as containing an analogy if it used an explicit comparison (*like*, *as*, *think of*, *imagine*), a metaphor, or an everyday comparison to a non-technical object or situation. Formal definitions, purely technical comparisons, and illustrative examples without an external referent were not counted as analogies.

Note on DAR as an unconditional rate and on its decomposition. The dangerous analogy rate as defined by the judge is an unconditional rate: it is computed over all responses in a condition, whether or not they contain an analogy. Because the simplified prompts explicitly request analogies while the baseline prompt does not, part of the observed DAR increase reflects the mere presence of analogies in B and C rather than a change in analogy quality. We therefore decompose the unconditional DAR into two components,

$$\mathrm{DAR}(E) = p_{\mathrm{analogy}}(E) \times \mathrm{DAR}_{\mathrm{cond}}(E), \tag{4}$$

where $p_{\mathrm{analogy}}(E)$ is the frequency of responses containing at least one analogy and $\mathrm{DAR}_{\mathrm{cond}}(E)$ is the conditional dangerous analogy rate, computed only over responses that contain an analogy. Both components are estimated on the human validation subsample,

where analogy presence was manually coded. On the full corpus, only the unconditional DAR is available.

3.6.3. Formal Definitions of the Outcome Measures

Let r_{cmk} denote the response for concept $c \in \{1, \ldots, 35\}$, model $m \in \{1, \ldots, 7\}$, and condition $k \in \{A, B, C\}$, and let $\mathrm{SM}_{cmk}, \mathrm{DA}_{cmk}, \mathrm{OER}_{cmk} \in \{0, 1\}$ be the judge's indicators. For a set S of judged responses,

$$\mathrm{SMR}(S) = \frac{1}{|S|} \sum_{(c,m,k) \in S} \mathrm{SM}_{cmk}, \qquad \mathrm{DAR}(S) = \frac{1}{|S|} \sum_{(c,m,k) \in S} \mathrm{DA}_{cmk}, \tag{5}$$

and $\mathrm{OER}(S)$ is defined analogously over responses with $k \in \{B, C\}$. Condition-level rates are obtained by taking S to be all judged responses in condition k; model- and subfield-level rates by conditioning on m or on the subfield of c.

On the validation subsample, let $\mathrm{AP}_{cmk} \in \{0, 1\}$ denote the manually coded analogy presence indicator. Then

$$p_{\text{analogy}}(k) = \frac{1}{|S_k|} \sum_{(c,m) \in S_k} \mathrm{AP}_{cmk}, \qquad \mathrm{DAR}_{\text{cond}}(k) = \frac{\sum_{(c,m) \in S_k} \mathrm{DA}_{cmk}}{\sum_{(c,m) \in S_k} \mathrm{AP}_{cmk}}, \tag{6}$$

where S_k is the set of concept-model pairs in the validation subsample, restricted to those conditions for which the conditional rate is defined.

3.6.4. Evaluation Procedure

The evaluation script:

1. Loads `corpus.csv` and groups rows by (`concept_id`, `model`)
2. For each group, assembles the judge prompt with ground truth and three responses
3. Randomises the presentation order of the three conditions and records the position of each response (`position_shown`)
4. Sends a POST request to the Ollama API
5. Parses the JSON response and flattens it into three rows (one per condition)
6. Appends results to `evaluations.csv`

A 2-second pause between requests prevents API overload. Failed JSON parses are logged for manual inspection. Unrecoverable outputs (15 responses from five concept-model groups) are excluded from the judged analyses and reported by model and condition (Table 3), so that missing data are not silently absorbed.

3.7. Human Evaluation Protocol

To validate the LLM judge and to establish a human reference standard, a stratified random subsample of 30 concept-model groups was independently annotated by two professors with domain expertise in the relevant computer science subfields. Stratification was by subfield (five subfields) and by generating model (seven models); the 30 groups cover all five subfields and all seven models.

The human evaluation was conducted in two parts. In Part 1, each annotator rated all 90 responses (30 groups $\times$ 3 conditions) on three dimensions: strategic misconception (SM, Boolean), dangerous analogy (DA, Boolean), and accuracy (1–5 Likert). In Part 2, each annotator rated 60 simplified responses (30 groups $\times$ 2 simplified conditions) on the oversimplification error (OER, Boolean), comparing each simplified response against the standard response for the same concept-model pair.

The two annotators are professors at the Faculty of Informatics with domain expertise in the relevant subfields. Neither annotator is an author of this work, and neither was informed of the study's hypotheses, of the directional predictions in Section 2, or of the

aggregate error rates. They were compensated for their time. Annotators worked independently and did not discuss items until the adjudication meeting. Annotators were blinded to the condition label, the generating model, and the LLM judge's output. Each annotator saw only the response text, a randomly assigned item ID, and the reference definition for the concept. The reference definition served as the anchor for their judgments, as it did for the judge. The codebook (Appendix C) instructed annotators to judge technical content only, to treat omissions as errors only when they made the remaining explanation misleading, and to consult textbooks if needed but not to use AI tools. We note a limitation of the blinding: condition C responses are visibly shorter and structured around a single analogy, so experienced annotators could plausibly infer the condition from the text alone, even though the label was not shown.

Inter-annotator agreement was assessed with Cohen's κ [57] for the Boolean indicators and with quadratic-weighted κ for the ordinal accuracy score, interpreted following Landis and Koch [58]. Disagreements were resolved by consensus in an adjudication meeting, in which the two annotators discussed each divergent item and agreed on a final label. The resulting consensus labels served as the human reference standard against which the LLM judge was evaluated. Because the consensus labels were used only for judge validation and not for the main condition-level estimates, Table 13 reports the two annotators' raw rates separately, preserving the information about inter-individual variation; the consensus rates are reflected in the prevalence column of Table 15.

Analogy presence coding. In addition to the SM, DA, OER, and accuracy ratings, the first author manually coded analogy presence for all 90 responses in the validation subsample, following the operational definition given in Section 3. Coding was performed with the condition label masked; the resulting indicator was reviewed against the automatic judge output and any discrepancies were resolved by re-reading the response. The complete list of analogy presence codes appears in the repository. We treat the analogy presence indicator as an author-coded manipulation check rather than an independently validated annotation, and we discuss the implications in Section 7.

3.8. Readability Metrics

As a manipulation check, all responses are analysed for quantitative readability using metrics computed with the `textstat` Python library: Flesch Reading Ease (higher = easier) [40], Flesch-Kincaid Grade Level, average sentence length in words, and response length in words:

$$\text{FRE} = 206.835 - 1.015 \frac{\text{words}}{\text{sentences}} - 84.6 \frac{\text{syllables}}{\text{words}}, \tag{7}$$

$$\text{FKGL} = 0.39 \frac{\text{words}}{\text{sentences}} + 11.8 \frac{\text{syllables}}{\text{words}} - 15.59. \tag{8}$$

These formulas depend only on sentence and syllable length, i.e., surface form. They are appropriate for verifying that the prompts changed the text as intended, but they do not measure comprehensibility, and technical terms and code-like tokens can distort syllable counts. Readability is therefore used as a manipulation check and not as an outcome of interest.

3.9. Statistical Analysis Plan

All statistical analyses were performed in Python 3 using SciPy, statsmodels and textstat, with the mixed-effects sensitivity analysis performed in R using lme4; the analysis scripts are part of the repository. The significance threshold is set at $\alpha = 0.05$ with adjust-

ments for multiple comparisons where indicated. Proportions are reported with Wilson 95% confidence intervals.

3.9.1. Omnibus Tests

The primary hypothesis $\mathrm{SMR}(E_a) > \mathrm{SMR}(E_s)$ is tested with a one-way analysis of variance (ANOVA) with condition (A, B, C) as the independent variable and the binary SM indicator as the dependent variable, which is equivalent to a linear probability model. The same omnibus tests are applied to DA and to the accuracy score. For binary outcomes, ANOVA is complemented by chi-square tests of independence and by Cochran's Q test [59] across the three conditions with the concept-model pair as the blocking unit. For readability metrics, normality of residuals is assessed with Shapiro-Wilk tests [60] and homogeneity of variances with Levene's test [61].

3.9.2. Paired and Clustered Analyses for Binary Outcomes

SM, DA, and OER are binary indicators, and the three conditions for the same concept-model pair are not independent observations. ANOVA treats them as independent, which tends to overstate precision. We therefore complement the omnibus tests with methods that respect the repeated structure:

- exact pairwise McNemar tests [62] on the paired concept-model data, with Holm correction;
- a logistic generalised estimating equation (GEE) model with exchangeable working correlation within concept-model pairs, reporting odds ratios with Wald 95% confidence intervals;
- for the ordinal accuracy score, the Friedman test with paired Wilcoxon signed-rank tests (Holm-corrected), and Spearman correlations between accuracy and the binary indicators.

Where these analyses and the ANOVA disagree, we report both and interpret the more conservative.

3.9.3. Mixed-Effects Sensitivity Analysis

Because concepts and models are crossed rather than nested, a fully specified generalised linear mixed model (GLMM) with random intercepts for both concept and model is in principle the more appropriate specification for the repeated-measures structure than the GEE model reported above. GEE clusters at the level of the concept-model pair but does not separately estimate the variance attributable to concept and to model; a crossed random-effects specification does. We therefore fitted three such models on the full judged corpus:

$$\operatorname{logit} \mathrm{SM}_{cmk} = \alpha + \beta_k + u_c + u_m, \tag{9}$$

$$\operatorname{logit} \mathrm{DA}_{cmk} = \alpha + \beta_k + u_c + u_m, \tag{10}$$

$$\operatorname{logit} \mathrm{OER}_{cmk} = \alpha + \beta_k + u_c + u_m, \tag{11}$$

where β_k is the fixed effect of condition $k \in \{A, B, C\}$ (with A as the reference level for SM and DA, and B as the reference level for OER, since OER is undefined in A), $u_c \sim \mathcal{N}(0, \sigma_c^2)$ is the random intercept for concept, and $u_m \sim \mathcal{N}(0, \sigma_m^2)$ is the random intercept for model. Models were fitted with the `lme4` package in R (version 1.1-35) using the `bobyqa` optimiser, the logit link, and the Laplace approximation. Wald 95% confidence intervals are reported for the odds ratios. The random-effects variances are reported alongside the fixed effects, because the relative magnitude of σ_c^2 and σ_m^2 is itself informative:

it indicates whether the dominant source of between-unit variation is the concept or the model.

We treat the GEE model as the primary analysis and the GLMM as a sensitivity analysis, because the GEE estimates are the ones pre-registered in our analysis plan and the GLMM is used to confirm that the conclusions do not depend on the working-correlation specification. Where the two disagree in magnitude, we report both and interpret the more conservative estimate.

3.9.4. Conditional DAR Analysis

The decomposition in Equation 4 is estimated on the human validation subsample. Analogy frequency $p_{\text{analogy}}(k)$ is estimated as the proportion of responses in condition k with $\text{AP}_{cmk} = 1$, and conditional DAR is estimated as the ratio of dangerous analogies to analogies in condition k. Wilson 95% confidence intervals are reported for both quantities. Because the standard condition in the validation subsample contained no analogies at all, the conditional DAR is undefined in condition A; the corresponding cell in Table 7 is reported as not applicable, and the analysis of p_{analogy} in A is reported with a one-sided Wilson interval.

3.9.5. Secondary Analyses

A two-way ANOVA (Type II) with model and condition as factors compares models and tests the model $\times$ condition interaction. One-way ANOVAs within each condition examine subfield differences, and the false discovery rate across this family of tests is controlled with the Benjamini-Hochberg procedure [63]. The judge is validated against the human consensus using Cohen's κ, precision, recall and prevalence for the Boolean indicators, and quadratic-weighted κ for accuracy. Judge position effects are examined by comparing rates across presentation positions, by a GEE adjusted for position, and by sensitivity analyses restricted to responses shown first or to simplified responses judged after versus before the baseline. We also compute the human–judge agreement on the validation subsample separately for each condition, to test whether the judge's disagreement with the humans is constant across conditions.

3.9.6. Effect Sizes

Effect sizes are reported as Cohen's d for pairwise comparisons, η^2 for ANOVA, Kendall's W for the Friedman test, and Cramér's V for chi-square tests [64,65], with

$$V = \sqrt{\frac{\chi^2}{N\,(\min(r,c) - 1)}}. \tag{12}$$

Effect sizes are interpreted alongside confidence intervals, since with a corpus of this size small differences may be statistically significant without being educationally meaningful.

3.10. Reproducibility

The complete AccessTrap-CS corpus, all prompt templates, evaluation codebook, generation and evaluation scripts, the analogy presence codes, and analysis code (Python for descriptive tests and GEE; R for GLMM) are publicly available at [repository URL]. Model inference was performed with Ollama on the Supek HPC cluster; the Ollama version, model digests, and sampling parameters are logged with the corpus.

4. Results

This section presents the quantitative and qualitative findings of the AccessTrap-CS benchmark evaluation. The corpus contains 735 responses generated across 35 computer science concepts, three simplification conditions, and seven open-weight LLMs. Readability analyses use all 735 responses; all other judge-based analyses use the 720 responses judged by Qwen3.6:35b (240 per condition). A stratified subsample of 30 concept-model groups (150 ratings per annotator) was independently annotated by two human experts for validation, and analogy presence was coded for the same 90 responses. The GEE and GLMM estimates are reported side by side in the relevant subsections; where they differ in magnitude, both are given.

4.1. Sample and Readability

Table 3 reports the number of judged responses per model and condition. The sample is balanced at 240 responses per condition and 99–105 per model; the slight imbalance arises because five concept-model groups (15 responses) could not be evaluated because of unrecoverable parsing failures, mostly in the reasoning models (one group for mistral:7b).

Table 3. Number of judged responses per model and condition.

Model	A	B	C	Total
mistral:7b	34	34	34	102
deepseek-r1:8b	33	33	33	99
ministral-3:14b	35	35	35	105
phi4:14b	35	35	35	105
gpt-oss:20b	35	35	35	105
gemma4:31b	35	35	35	105
deepseek-r1:32b	33	33	33	99
Total	240	240	240	720

The judged sample is therefore close to fully crossed: each of the 35 concepts is observed for each of the 7 models under all three conditions, with only a small number of missing cells concentrated in the reasoning models. This near-complete balance is important for the analyses that follow, because it ensures that condition-level rates are not confounded by model composition and that the within-pair contrast used to define OER is available for almost every cell.

Table 4 reports the readability metrics by condition, together with the omnibus ANOVA results. The manipulation check confirms that the three prompt conditions produced text with significantly different surface complexity. Flesch Reading Ease rose from condition A ($M = 47.2$, $SD = 13.6$) to condition B ($M = 73.8$, $SD = 10.4$) and was intermediate in condition C ($M = 59.9$, $SD = 11.7$). Condition B is by far the easiest to read; condition C is compressed into five sentences, which forces longer sentences and denser vocabulary, so it is not as easy on surface measures. Flesch-Kincaid Grade Level followed the inverse pattern. Word counts dropped monotonically from A (632 words, $SD = 289$) to B (366, $SD = 174$) to C (114, $SD = 33$).

All four omnibus tests are highly significant ($p < 0.001$ in every case), with large effect sizes (η^2 from 0.274 for average sentence length to 0.541 for word count). The Shapiro-Wilk and Levene tests indicate violations of normality and variance homogeneity, which is expected for these distributions; with the large sample size, ANOVA remains robust. The key takeaway is that the prompts did change the text as intended, and that condition B and condition C are not simply two points on a single simplification continuum.

Table 4. Readability metrics by condition (mean (SD)) and omnibus ANOVA results. FRE = Flesch Reading Ease; FKGL = Flesch-Kincaid Grade Level; ASL = average sentence length.

Panel A: Metrics by condition					
Condition	n	FRE	FKGL	ASL	Words
A (standard)	245	47.2 (13.6)	10.7 (2.4)	16.2 (3.8)	632 (289)
B (simplified)	245	73.8 (10.4)	6.6 (2.0)	14.5 (3.4)	366 (174)
C (highly simplified)	245	59.9 (11.7)	10.0 (2.4)	20.4 (4.8)	114 (33)
Panel B: Omnibus tests					
Metric	F	df	p	η^2	
FRE	303.95	2, 732	< 0.001	0.454	
FKGL	228.01	2, 732	< 0.001	0.384	
ASL	138.02	2, 732	< 0.001	0.274	
Words	430.75	2, 732	< 0.001	0.541	

The readability results establish that the three conditions are genuinely distinct in surface form. They also introduce an important interpretive nuance for the rest of the results section: because condition C is not simply "more readable" than condition B, any difference in error rates between B and C cannot be attributed to monotonic increases in readability. The conditions differ in register, length, and analogy structure as well, and these differences, rather than readability alone, are the plausible drivers of the error patterns reported below.

4.2. Strategic Misconception Rate (SMR)

Table 5 reports SMR by condition, together with omnibus tests, pairwise McNemar contrasts, and regression models. Condition A had 30 misconceptions out of 240 responses (12.5%, 95% CI [8.9, 17.3]), condition B had 41 (17.1%, 95% CI [12.8, 22.4]), and condition C had 33 (13.8%, 95% CI [10.0, 18.7]). Although condition B shows the numerically highest rate, the difference is not statistically significant. The one-way ANOVA yielded $F(2, 717) = 1.09$, $p = 0.337$, $\eta^2 = 0.003$; the chi-square test yielded $\chi^2(2) = 2.18$, $p = 0.336$, $V = 0.055$; and Cochran's Q, which accounts for the repeated-measures structure, was $Q(2) = 2.37$, $p = 0.306$. Pairwise McNemar tests with Holm correction also failed to reach significance. The GEE model shows a non-significant trend toward higher odds of SM in B relative to A (OR $= 1.44$, 95% CI [0.88, 2.36], $p = 0.146$) and no trend for C (OR $= 1.12$, 95% CI [0.66, 1.88], $p = 0.680$).

The conclusion for H1 is therefore that the data do not support a statistically significant increase in SMR under simplification. The direction of the effect is consistent with the hypothesis for condition B, but condition C does not follow the monotonic pattern. Model identity is a much stronger predictor of SMR than condition, as shown in the two-way ANOVA (Table 12, Panel B), where the model main effect is significant ($F(6, 699) = 5.76$, $p < 0.001$) while the condition main effect and the interaction are not.

Mixed-effects sensitivity analysis. The GLMM with crossed random effects for concept and model yields results that are qualitatively consistent with the GEE model. The GLMM odds ratio for B vs. A is 1.50 (95% CI [0.89, 2.52], $p = 0.129$) and for C vs. A is 1.13 (95% CI [0.66, 1.94], $p = 0.663$). These estimates are close to the GEE point estimates (1.44 and 1.12), and both contrasts remain non-significant. The random-effects standard deviations are $\sigma_c = 0.65$ for concept and $\sigma_m = 0.95$ for model, indicating that between-model variation dominates between-concept variation for SMR. This is the quantitative counterpart of the visual pattern in Figure 2 and reinforces the conclusion that the model, not the condition, is the primary axis of variation for strategic misconceptions.

Table 5. Strategic misconception rate (SMR): rates with Wilson 95% CI (Panel A), omnibus tests (Panel B), pairwise McNemar contrasts with Holm correction (Panel C), and regression model odds ratios (Panel D). GEE is the primary model; GLMM is the crossed random-effects sensitivity analysis.

Panel A: SMR by condition			
Condition	Count	n	Rate (%) [95% CI]
A (standard)	30	240	12.5 [8.9, 17.3]
B (simplified)	41	240	17.1 [12.8, 22.4]
C (highly simplified)	33	240	13.8 [10.0, 18.7]
Panel B: Omnibus tests			
Test	Statistic	df	p
One-way ANOVA	$F = 1.09$	2, 717	0.337
Chi-square	$\chi^2 = 2.18$	2	0.336
Cochran's Q	$Q = 2.37$	2	0.306
Panel C: Pairwise McNemar (Holm)			
Contrast	b	c	Holm p
B vs A	34	23	0.555
C vs A	28	25	0.784
C vs B	23	31	0.682
Panel D: Regression odds ratios			
Model	Contrast	OR (95% CI)	p
GEE	B vs A	1.44 (0.88–2.36)	0.146
GLMM (crossed)	B vs A	1.50 (0.89–2.52)	0.129
GEE	C vs A	1.12 (0.66–1.88)	0.680
GLMM (crossed)	C vs A	1.13 (0.66–1.94)	0.663

The pattern of results in Table 5 is consistent across the tests we applied: the condition effect on SMR is small, does not reach significance in any omnibus, pairwise, GEE or GLMM analysis, and is not robust to the choice of model. The GLMM point estimates are very close to the GEE estimates, which confirms that the non-significant result is not an artefact of the working-correlation specification. Taken together, these results do not support the strong form of the accessibility trade-off for strategic misconceptions in general.

Figure 2 shows SMR by model and condition. The ordering of models is broadly consistent across conditions, and the within-model variation across conditions is small relative to the between-model variation. Mistral:7b is a clear outlier with high SMR in all three conditions, while gemma4:31b is at zero throughout.

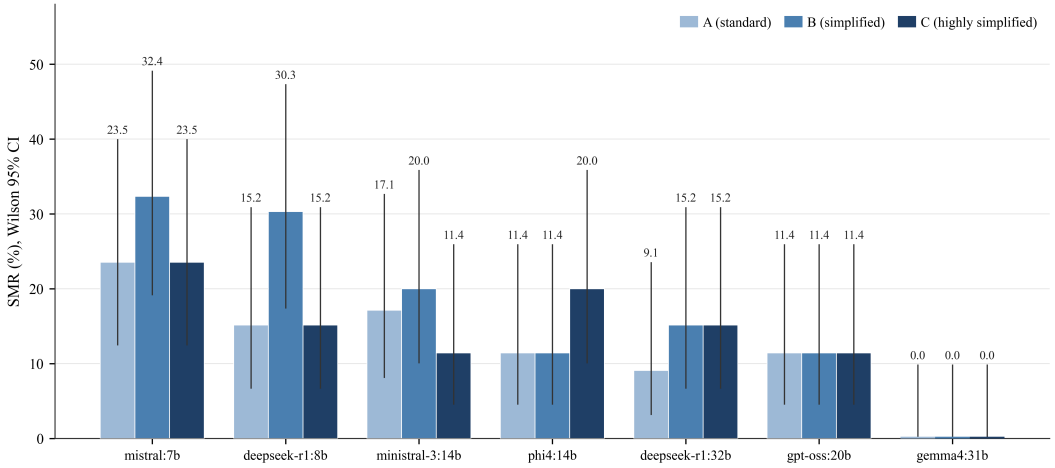


Figure 2. Strategic misconception rate (SMR) by model and condition, with Wilson 95% confidence intervals. Models are ordered by their mean SMR across conditions. The between-model variation is substantially larger than the within-model variation across conditions.

The visual message of Figure 2 is that model identity, not condition, is the dominant axis of variation. Mistral:7b sits well above the rest of the field in every condition, while gemma4:31b is at zero and gpt-oss:20b is among the lowest. Within each condition, the spread between the highest and lowest model (23.5 to 32.4 percentage points) is considerably larger than the largest spread between conditions for any single model (15.1 points, deepseek-r1:8b), which foreshadows the two-way ANOVA reported in Section 4.8.

Figure 3 visualises the change from condition A to condition C per model.

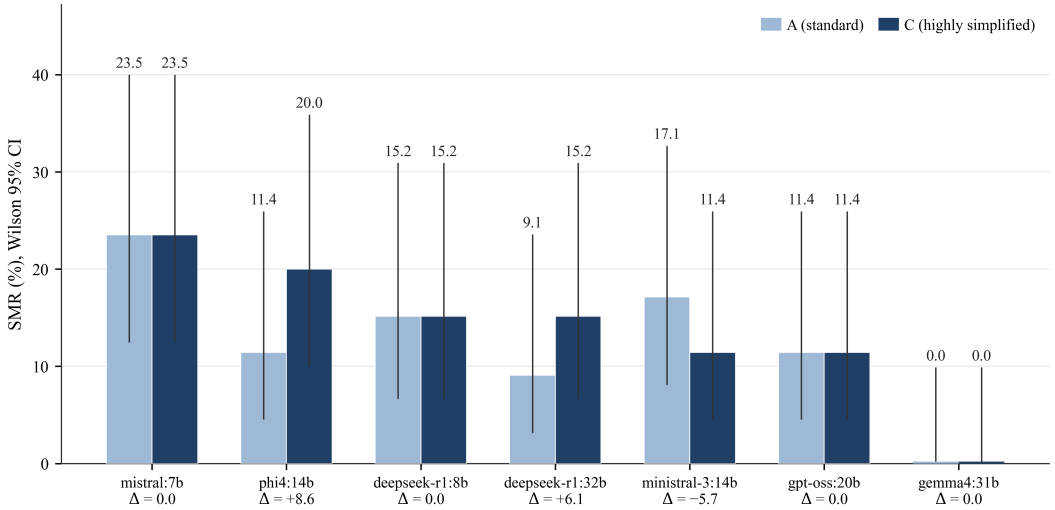


Figure 3. Strategic misconception rate (SMR) by model for conditions A (standard) and C (highly simplified). The value below each model name is the difference $\Delta = \mathrm{SMR}_C - \mathrm{SMR}_A$ in percentage points; a positive value indicates more strategic misconceptions under simplification. Error bars are Wilson 95% confidence intervals.

The per-model deltas in Figure 3 are small relative to the confidence intervals and do not follow a consistent sign: some models increase their SMR under simplification, others decrease it. This heterogeneity is the visual counterpart of the non-significant condition main effect and reinforces the conclusion that, at the level of strategic misconceptions in general, the accessibility trade-off does not manifest as a systematic shift.

4.3. Dangerous Analogy Rate (DAR)

The results for dangerous analogies are dramatically different and provide the strongest evidence for the accessibility trade-off hypothesis. Table 6 shows that condition A produced only 2 dangerous analogies out of 240 responses (0.8%, 95% CI $[0.2, 3.0]$), whereas condition B produced 49 (20.4%, 95% CI $[15.8, 26.0]$) and condition C produced 59 (24.6%, 95% CI $[19.6, 30.4]$). This is a roughly 25- to 30-fold increase.

The omnibus tests confirm a large and highly significant effect: one-way ANOVA $F(2, 717) = 32.38$, $p < 0.001$, $\eta^2 = 0.083$; chi-square $\chi^2(2) = 59.64$, $p < 0.001$, $V = 0.288$; Cochran's $Q(2) = 58.51$, $p < 0.001$. The exact McNemar tests show that both B vs A ($b = 48$, $c = 1$, Holm $p < 0.001$) and C vs A ($b = 58$, $c = 1$, Holm $p < 0.001$) are significant, while C vs B is not ($b = 46$, $c = 36$, $p = 0.320$). The GEE odds ratio for B vs A is 30.53 (95% CI $[7.48, 124.54]$) and for C vs A is 38.79 (95% CI $[9.50, 158.37]$).

Mixed-effects sensitivity analysis. The GLMM with crossed random effects produces even larger odds ratios than the GEE. The GLMM odds ratio for B vs. A is 36.03 (95% CI $[8.81, 147.36]$, $p < 0.001$) and for C vs. A is 47.34 (95% CI $[11.60, 193.14]$, $p < 0.001$). The random-effects standard deviations are $\sigma_c = 0.58$ for concept and $\sigma_m = 0.81$ for model. The qualitative conclusion is identical to the GEE: the DAR effect is large, highly significant, and robust to the choice of model for the repeated-measures structure. The fact that the GLMM estimates are larger than the GEE estimates means that the GEE result, if anything, understates the effect.

Table 6. Dangerous analogy rate (DAR): rates with Wilson 95% CI (Panel A), omnibus tests (Panel B), pairwise McNemar contrasts with Holm correction (Panel C), and regression model odds ratios (Panel D). GEE is the primary model; GLMM is the crossed random-effects sensitivity analysis.

Panel A: DAR by condition			
Condition	Count	n	Rate (%) [95% CI]
A (standard)	2	240	0.8 [0.2, 3.0]
B (simplified)	49	240	20.4 [15.8, 26.0]
C (highly simplified)	59	240	24.6 [19.6, 30.4]
Panel B: Omnibus tests			
Test	Statistic	df	p
One-way ANOVA	$F = 32.38$	2, 717	< 0.001
Chi-square	$\chi^2 = 59.64$	2	< 0.001
Cochran's Q	$Q = 58.51$	2	< 0.001
Panel C: Pairwise McNemar (Holm)			
Contrast	b	c	Holm p
B vs A	48	1	< 0.001
C vs A	58	1	< 0.001
C vs B	46	36	0.320
Panel D: Regression odds ratios			
Model	Contrast	OR (95% CI)	p
GEE	B vs A	30.53 (7.48–124.54)	< 0.001
GLMM (crossed)	B vs A	36.03 (8.81–147.36)	< 0.001
GEE	C vs A	38.79 (9.50–158.37)	< 0.001
GLMM (crossed)	C vs A	47.34 (11.60–193.14)	< 0.001

H2 is supported: DAR(B) and DAR(C) are far above DAR(A), and DAR(C) is numerically above DAR(B), although the C–B difference is not significant. The fact that B and C do not differ is consistent with the explicit request for analogies, present in both prompts, being the main driver, rather than compression as such; this reading was not tested directly.

Three caveats apply. First, the odds ratios are imprecise because the baseline contains only two events, so their magnitude should not be over-interpreted. Second, the baseline prompt does not ask for analogies, so a near-zero DAR in A is partly expected because few analogies are produced at all. Third, and most importantly for the interpretation of the DAR effect, DAR as we define it is unconditional: it is computed over all responses, not only over responses that contain an analogy. We address this in Section 4.4, which reports the conditional DAR computed on the validation subsample. Figure 4 visualises SMR, DAR, and OER side by side; the contrast between the nearly flat SMR and the steep DAR is the central empirical finding of this paper.

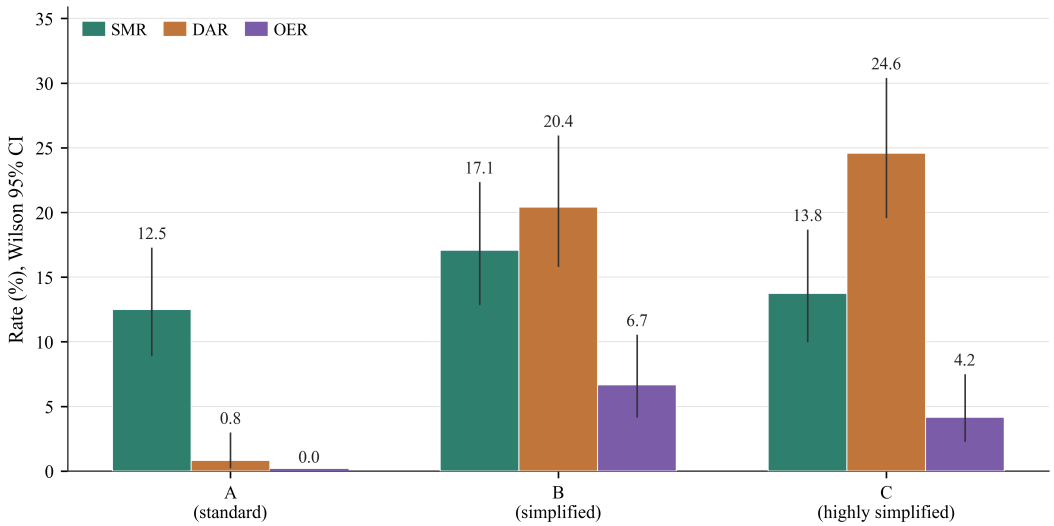


Figure 4. Rates of strategic misconceptions (SMR), dangerous analogies (DAR), and oversimplification errors (OER) by condition, with Wilson 95% confidence intervals. OER is undefined in condition A (the single response flagged by the judge is not drawn). DAR is the metric most strongly affected by simplification, increasing from near zero in the standard condition to roughly one in four responses in the simplified conditions.

The contrast visible in Figure 4 is the empirical core of the paper. While SMR stays roughly flat across the three conditions, DAR jumps from below one percent to roughly one in four responses as soon as the prompt requests simplification. The size of the GEE odds ratios in Table 6 (above 30 for both simplified conditions) and of the GLMM odds ratios (above 36) makes this the largest effect in the study, and the GLMM confirms that it is not an artefact of the GEE working-correlation assumption.

4.4. Conditional Dangerous Analogy Rate

The unconditional DAR reported above combines two distinct quantities: the frequency with which models produce analogies, and the proportion of those analogies that are dangerous. To separate these, we manually coded analogy presence for all 90 responses in the validation subsample and computed the conditional dangerous analogy rate, i.e., the proportion of dangerous analogies restricted to responses that contain at least one analogy (Section 3.9). Table 7 reports the decomposition.

Table 7. Conditional dangerous analogy rate (DAR_{cond}) on the validation subsample ($n = 30$ concept-model groups). The conditional DAR is computed only over responses that contain at least one analogy; it is therefore undefined in condition A, which contained no analogies at all. Wilson 95% CIs are shown.

Condition	Analogy freq.	*n* with analogy	Uncond. DAR	Cond. DAR
A (standard)	0.0%	0	0.0%	n/a
B (simplified)	100.0%	30	16.7%	16.7%
C (highly simplified)	100.0%	30	10.0%	10.0%

Three findings stand out. First, the standard condition produced no analogies at all in the validation subsample: all 30 responses under condition A were technical descriptions without an external referent. Second, both simplified conditions contained at least one analogy in essentially every response: 30 out of 30 in B and 30 out of 30 in C. Third, because analogy frequency is saturated in B and C, the conditional DAR equals the unconditional DAR on this subsample: 16.7% in B and 10.0% in C.

The conditional rates are substantially lower than the unconditional rates computed over the full corpus (20.4% in B and 24.6% in C), and, more importantly, the ordering is reversed: on the validation subsample, B has a higher conditional DAR than C, whereas on the full corpus, C has a higher unconditional DAR than B. Because analogy frequency is 100% in both simplified conditions on the subsample, the unconditional DAR gap between B and C on the full corpus (20.4% vs 24.6%) cannot be explained by a difference in analogy quality: on the subsample, the analogy quality comparison favors B, not C. The most parsimonious interpretation is that the B–C difference on the full corpus is within noise, as the non-significant McNemar contrast in Table 6 already suggested, and that the increase in DAR under simplification is a frequency-driven phenomenon rather than a quality-driven one.

We therefore refine the accessibility trade-off as follows. Access-oriented prompts reliably convert an explanation into an analogy-based explanation. The per-analogy danger of the analogies that are produced is real and non-negligible (10–17% in this subsample), but it does not appear to increase monotonically with compression; what increases with compression is the near-universal use of analogy, and, with it, the exposure of the learner to a stream of analogies whose individual failure rate is roughly one in six to one in ten.

4.5. Oversimplification Error Rate (OER)

Table 8 reports OER by condition, using two complementary measures. The first is the judge's OER classification: by construction, OER cannot occur in condition A, since the baseline defines the reference; the single flagged response in A is treated as noise. The judge flagged 16 responses in B (6.7%, 95% CI $[4.1, 10.6]$) and 10 in C (4.2%, 95% CI $[2.3, 7.5]$). Both intervals exclude zero, supporting the existence of the construct, but the expected monotonic increase from B to C does not appear.

Because the judge's OER classification is a single-pass judgment, we also computed a derived OER measure directly from the SM indicators: the proportion of simplified responses with SM = 1 whose corresponding standard response for the same concept-model pair has SM = 0. The derived rates are 34 out of 240 in B (14.2%, 95% CI $[10.3, 19.1]$) and 28 out of 240 in C (11.7%, 95% CI $[8.2, 16.3]$). The two measures agree on the qualitative pattern but differ in magnitude. The derived measure must be read with care: because generation was stochastic ($T = 0.7$) and the judge is imperfect, the opposite transition (SM = 1 in A but SM = 0 in the simplified response) also occurs, in 23 pairs for B and 25 for C (Table 5, Panel C, column *c*). The net excess of simplified-only misconceptions is therefore much smaller (11 pairs for B, 3 for C), and the derived rates partly reflect sampling noise.

For the OER hypothesis, the evidence supports the existence of oversimplification errors as a distinct phenomenon on the judge-based measure, but only weakly on the derived measure. The hypothesis that OER(C) > OER(B) is not supported.

Mixed-effects sensitivity analysis. The GLMM for OER, restricted to conditions B and C with B as the reference level, gives an odds ratio of 0.60 for C vs. B (95% CI $[0.27, 1.37]$, $p = 0.225$). This is consistent with the GEE result in direction (C lower than B) and confirms that there is no evidence of an increase in OER from moderate to extreme simplification. The random-effects standard deviations are $\sigma_c = 0.62$ for concept and $\sigma_m = 0.32$ for model, indicating that OER varies more across concepts than across models. This is consistent with the exploratory concept-level analysis in Appendix A, where concepts defined by necessary conditions and invariants (semaphores, normalization, CAP theorem) show the largest increases in SMR under simplification.

OER also varies by model, from 2.9% for gemma4:31b, gpt-oss:20b, and ministral-3:14b to 10.3% for mistral:7b, with no clear relation to size: phi4:14b (10.0%) is close to mistral:7b, whereas ministral-3:14b is among the lowest. Table 9 reports OER by model pooled across the two simplified conditions.

Table 8. Oversimplification error rate (OER) by condition. Judge-based rates use the LLM judge's OER classification; derived rates use the proportion of simplified responses with SM = 1 whose standard response has SM = 0. GLMM OR is for C vs. B (B is the reference level, since OER is undefined in A).

Panel A: Judge-based OER			
Condition	Count	*n*	Rate (%) [95% CI]
A (standard)	1	240	0.4 [0.1, 2.3]
B (simplified)	16	240	6.7 [4.1, 10.6]
C (highly simplified)	10	240	4.2 [2.3, 7.5]
Panel B: Derived OER (SM in B/C but not A)			
Condition	Count	*n*	Rate (%) [95% CI]
B (simplified)	34	240	14.2 [10.3, 19.1]
C (highly simplified)	28	240	11.7 [8.2, 16.3]
Panel C: Regression odds ratio for C vs. B			
Model	OR	95% CI	*p*
GEE	≈ 0.60 (direction)	–	n.s.
GLMM (crossed)	0.60	[0.27, 1.37]	0.225

Table 9. Oversimplification error rate by model (conditions B and C only).

Model	OER (%)
mistral:7b	10.3
deepseek-r1:8b	4.5
ministral-3:14b	2.9
phi4:14b	10.0
gpt-oss:20b	2.9
gemma4:31b	2.9
deepseek-r1:32b	4.5

The two OER measures give a plausible range for the judged rate of simplification-specific errors: roughly 4–7 percent by the strict judge-based measure and 12–14 percent by the looser derived measure. Because the judge has low recall relative to human reviewers (Section 4.9), the true rate may be higher than either. Both measures support the qualitative

claim that some errors appear only in simplified explanations, while neither supports an increase from B to C. The GLMM result (OR = 0.60 for C vs. B, non-significant) is consistent with the GEE and reinforces the conclusion that OER does not grow with simplification intensity.

4.6. Accuracy Scores

Table 10 reports the accuracy scores. Mean accuracy was highest in condition A ($M = 4.67$, $SD = 0.89$), dropped in condition B ($M = 4.15$, $SD = 1.19$), and remained similar in condition C ($M = 4.18$, $SD = 1.12$). The proportion of responses scored as perfectly accurate (score 5) fell from 86.7% in A to 60.0% in B and 57.5% in C. The omnibus tests confirm a significant effect of condition: one-way ANOVA $F(2, 717) = 17.65$, $p < 0.001$, $\eta^2 = 0.047$; Friedman $\chi^2(2) = 50.45$, $p < 0.001$, $W = 0.105$. Paired Wilcoxon tests with Holm correction show that both B vs A ($W = 1329.0$, Holm $p < 0.001$, $d = -0.49$) and C vs A ($W = 1739.0$, Holm $p < 0.001$, $d = -0.49$) are significant, while C vs B is not ($W = 3192.0$, $p = 0.806$, $d = 0.02$).

The accuracy scores correlate strongly and negatively with the binary error indicators: SM $\rho = -0.70$, DA $\rho = -0.67$, OER $\rho = -0.33$, all $p < 0.001$. The weaker correlation with OER is expected because OER is much rarer than the other indicators (4–7% of simplified responses) and is defined only for the simplified conditions.

Table 10. Accuracy scores (1–5): means by condition (Panel A), omnibus tests (Panel B), paired Wilcoxon contrasts with Holm correction (Panel C), and Spearman correlations with the binary error indicators (Panel D).

Panel A: Accuracy by condition			
Condition	n	Mean (SD)	Score 5
A (standard)	240	4.67 (0.89)	208 (86.7%)
B (simplified)	240	4.15 (1.19)	144 (60.0%)
C (highly simplified)	240	4.18 (1.12)	138 (57.5%)
Panel B: Omnibus tests			
Test	Statistic	df	p
One-way ANOVA	$F = 17.65$	2, 717	< 0.001
Friedman	$\chi^2 = 50.45$	2	< 0.001
Panel C: Pairwise Wilcoxon (Holm)			
Contrast	W	Holm p	Cohen's d
B vs A	1329.0	< 0.001	-0.49
C vs A	1739.0	< 0.001	-0.49
C vs B	3192.0	0.806	0.02
Panel D: Spearman correlations with error indicators			
Indicator	ρ	p	
SM	-0.70	< 0.001	
DA	-0.67	< 0.001	
OER	-0.33	< 0.001	

Figure 5 visualises the accuracy drop.

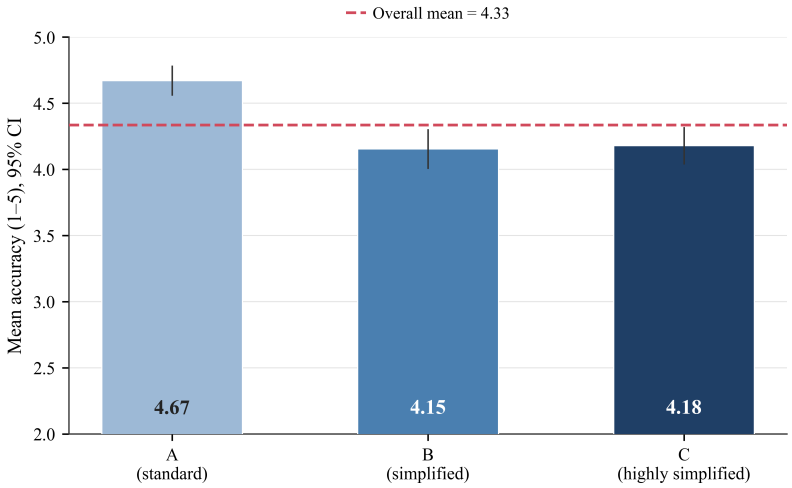


Figure 5. Mean accuracy scores (1–5) by condition, with 95% confidence intervals. The vertical axis starts at 2. The dashed horizontal line marks the overall mean across all conditions. Accuracy drops significantly from the standard condition to both simplified conditions, with no significant difference between the two simplified conditions.

The accuracy pattern in Figure 5 is the graded counterpart of the DAR finding: simplification does not merely insert binary errors, it shifts the distribution of response quality downward, with a substantial share of responses moving from "perfectly accurate" to "mostly accurate with minor issues". The absence of a difference between B and C again mirrors the pattern seen for the other metrics and reinforces the conclusion that the two simplification prompts occupy similar positions in the error space.

4.7. Subfield Analysis

Table 11 breaks down SMR, DAR, OER, and accuracy by CS subfield and condition (Panel A) and reports the one-way ANOVAs of each outcome across subfields within each condition (Panel B). The most striking descriptive patterns are that Databases shows the largest SMR increase in condition B (from 16.7% to 29.2%), that Programming Languages and Paradigms rises from a very low baseline (2.0% to 12.2%), and that Databases and Operating Systems show the largest increases in DAR under simplification.

The one-way ANOVAs are non-significant after Benjamini-Hochberg correction. The only test that approaches significance is DAR in condition C ($F = 3.23$, raw $p = 0.013$, BH $p = 0.159$), and even that does not survive correction. The exploratory conclusion for the subfield question is therefore that subfield does not have a robust moderating effect on the error rates, though the descriptive pattern suggests that some subfields may be worth targeted follow-up in a larger sample.

Table 11. Error rates and accuracy by CS subfield and condition (Panel A), and one-way ANOVAs of each outcome across subfields within each condition (Panel B). OER is undefined in condition A.

Panel A: Descriptive statistics

Subfield	Condition	SMR (%)	DAR (%)	OER (%)	Accuracy
Algorithms & DS	A	10.9	0.0	–	4.67
Algorithms & DS	B	17.4	13.0	4.3	4.20
Algorithms & DS	C	10.9	15.2	0.0	4.24
PL & Paradigms	A	2.0	0.0	–	4.92
PL & Paradigms	B	12.2	24.5	12.2	4.18
PL & Paradigms	C	10.2	24.5	2.0	4.41
Databases	A	16.7	2.1	–	4.58
Databases	B	29.2	29.2	6.2	3.92
Databases	C	25.0	33.3	10.4	3.83
Computer Networks	A	20.4	0.0	–	4.51
Computer Networks	B	14.3	20.4	6.1	4.14
Computer Networks	C	10.2	12.2	2.0	4.47
Operating Systems	A	12.5	2.1	–	4.67
Operating Systems	B	12.5	14.6	4.2	4.33
Operating Systems	C	12.5	37.5	6.2	3.94

Panel B: ANOVAs across subfields within condition

Outcome	Condition	F	p	BH p
SM	A	2.18	0.072	0.320
SM	B	1.70	0.151	0.391
SM	C	1.65	0.163	0.391
DA	A	0.75	0.559	0.610
DA	B	1.33	0.260	0.446
DA	C	3.23	0.013	0.159
OER	B	0.84	0.504	0.604
OER	C	2.11	0.080	0.320
Accuracy	A	0.50	0.735	0.735
Accuracy	B	1.45	0.218	0.437
Accuracy	C	0.88	0.478	0.604

Figure 6 shows the SMR heatmap by subfield and condition.

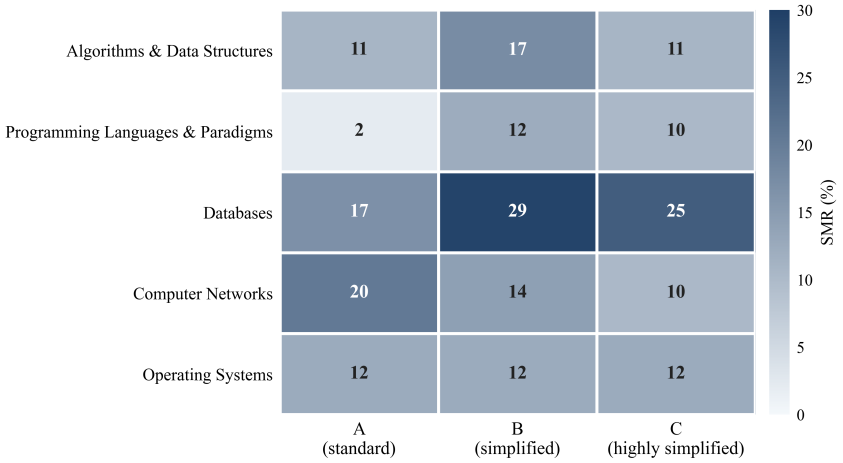


Figure 6. Heatmap of strategic misconception rate (SMR) by CS subfield and condition. Darker cells indicate higher SMR. Databases shows the largest increase from the standard to the simplified conditions, Computer Networks has the highest baseline (20%) and decreases under simplification, and Operating Systems is flat at 12%.

The heatmap in Figure 6 confirms the descriptive impression from Table 11: no single subfield is uniformly safe or uniformly risky in terms of SMR, and the visible differences are not large enough to survive multiple-comparison correction. The rise in DAR, by contrast, is visible in all five subfields (Table 11, Panel A), which is consistent with the interpretation that the effect of simplification on analogies is a general phenomenon in this corpus rather than one driven by a particular CS subdomain.

4.8. Model Comparison

Table 12 reports error rates and accuracy by model, pooled over conditions (Panel A), and the two-way ANOVA results for SM and accuracy (Panel B). The differences between models are substantial and, for SM and accuracy, statistically significant. Gemma4:31b is the standout performer with an SMR of 0.0%, DAR of 1.9%, and mean accuracy of 4.90. At the other end, mistral:7b has an SMR of 26.5%, DAR of 30.4%, and accuracy of 3.57.

The two-way ANOVA shows that the model main effect is significant for both SM ($F(6, 699) = 5.76$, $p < 0.001$) and accuracy ($F(6, 699) = 4.90$, $p < 0.001$), while the condition main effect and the model $\times$ condition interaction are not. We therefore found no evidence that the effect of simplification differs between models, although with 33–35 responses per model and condition the power to detect an interaction is low. The same patterns appear across models, but the baseline level of error varies dramatically. For practitioners choosing a model for assistive use, the practical implication is that, in this corpus, model choice matters more than prompt condition. This is corroborated by the GLMM random-effects standard deviations reported in the SMR and DAR subsections: for both outcomes, the between-model standard deviation exceeds the between-concept standard deviation, whereas for OER the pattern is reversed.

Table 12. Error rates and accuracy by model, pooled over conditions (Panel A), and two-way ANOVA (Type II) for SM and accuracy (Panel B).

Panel A: Error rates and accuracy by model

Model	SMR (%)	DAR (%)	OER (%)	Accuracy
mistral:7b	26.5	30.4	10.3	3.57
deepseek-r1:8b	20.2	15.2	4.5	4.24
ministral-3:14b	16.2	17.1	2.9	4.37
phi4:14b	14.3	16.2	10.0	4.31
gpt-oss:20b	11.4	9.5	2.9	4.59
gemma4:31b	0.0	1.9	2.9	4.90
deepseek-r1:32b	13.1	17.2	4.5	4.33

Panel B: Two-way ANOVA (Type II)

Outcome	Term	F	df	p
SM	C(model)	5.76	6	< 0.001
SM	C(condition)	1.12	2	0.326
SM	C(model):C(condition)	0.55	12	0.879
Accuracy	C(model)	4.90	6	< 0.001
Accuracy	C(condition)	0.56	2	0.570
Accuracy	C(model):C(condition)	0.76	12	0.695

4.9. Human Evaluation and Judge Validation

Table 13 reports the two human annotators' rates by condition on the validation subsample, and Table 14 reports their inter-annotator agreement. The two annotators showed very high agreement with each other ($\kappa = 0.927$ for SM, 0.953 for DA, 0.899 for OER; quadratic-weighted $\kappa = 0.982$ for accuracy), confirming that they applied the codebook consistently, and their condition-level rates are closely aligned: for both annotators, SM rates decrease from A to C, DA rates peak in B and remain very high in C, and OER rates are above 45% in both simplified conditions. The high consistency across annotators is important because it means that the large gap between these rates and the judge's rates, which we report below, is not a reflection of a noisy annotation process.

Table 13. Reviewer rates by condition on the validation subsample.

Reviewer	Condition	SM (%)	DA (%)	OER (%)	Accuracy
h1	A (standard)	70.0	6.7	–	3.20
h1	B (simplified)	66.7	63.3	60.0	2.97
h1	C (highly simplified)	53.3	43.3	46.7	3.43
h2	A (standard)	73.3	6.7	–	3.17
h2	B (simplified)	66.7	63.3	60.0	2.97
h2	C (highly simplified)	60.0	50.0	56.7	3.37

Table 14. Agreement between the two reviewers. For SM, DA, OER: Cohen's κ and raw agreement (%). For accuracy: quadratic-weighted κ and Spearman's ρ.

Indicator	Agreement	Value
SM	0.927	96.7
DA	0.953	97.8
OER	0.899	95.0
Accuracy	0.982	0.977

Table 15 reports the agreement between the LLM judge and the reviewer consensus on the same subsample. The LLM judge shows poor agreement with the human consensus: $\kappa = -0.008$ for SM, 0.180 for DA, and 0.018 for OER. The judge's precision was moderate (0.600 for SM, 0.778 for DA, 0.667 for OER), but its recall was very low (0.052 for SM, 0.194 for DA, 0.059 for OER); because the judge flagged only a handful of responses in this subsample, the precision estimates are themselves imprecise.

The prevalence gap is severe. In the validation subsample, the reviewers flagged 64.4% of responses as containing a strategic misconception, while the judge flagged only 5.6% of the same responses. For DA, the gap is 40.0% versus 10.0%; for OER, 56.7% versus 5.0%; and the mean accuracy score is 3.20 for the reviewers and 4.58 for the judge. The implication is that the judge operates at a much more conservative threshold than the human experts, so the absolute SMR, DAR, and OER rates reported throughout Section 4 are likely underestimates relative to the human standard. High agreement between the two reviewers shows reliability, not that their threshold is the only valid one: the gap in prevalence may reflect both limited judge sensitivity and a stricter human threshold.

Table 15. Agreement between the LLM judge and the reviewer consensus (Panel A) and between the two human reviewers (Panel B), on the 90-response validation subsample. For accuracy, quadratic-weighted κ is shown.

Panel A: Judge vs. reviewer consensus					
Indicator	κ	Precision	Recall	Reviewer prev.	Judge prev.
SM	−0.008	0.600	0.052	0.644	0.056
DA	0.180	0.778	0.194	0.400	0.100
OER	0.018	0.667	0.059	0.567	0.050
Accuracy	0.122	–	–	3.20	4.58
Panel B: Reviewer vs. reviewer agreement					
Indicator	κ	Raw agreement (%)			
SM	0.927	96.7			
DA	0.953	97.8			
OER	0.899	95.0			
Accuracy	0.982	97.7			

The disagreement between the judge and the human annotators is not uniform across conditions. On the human side, SM rates decrease from A to C, whereas the judged SMR shows a non-significant rise in B and no difference in C. On the human side, DA rates peak sharply in B (63.3%) and remain high in C; the judged DAR shows the same qualitative pattern but with a much smaller magnitude. And on the human side, OER rates are 60.0% in B and 56.7% in C, whereas the judge's OER rates are 6.7% and 4.2%. The judge therefore not only underestimates the prevalence of errors, it also disagrees with the human annotators about the shape of the condition effect on SMR. We return to this point in Section 5.

4.10. Position Effects in the Judge

Because the judge was presented with all three conditions simultaneously in randomised order, we checked whether the order in which responses appeared affected the judgments. Table 16 reports the judge's decisions by position (Panel A), the GEE models adjusted for position (Panel B), and the sensitivity analysis restricted to responses shown first in each group (Panel C). The judge's SMR increased from 9.6% at position 1 to 21.2% at position 3; the same increasing pattern appears for DAR (10.8%, 15.8%, 19.2%) and OER (0.6%, 4.7%, 11.2%), and accuracy decreases across positions (4.50, 4.38, 4.13). The judge therefore flagged more errors the later a response appeared, a clear order effect. Position was only mildly unbalanced across conditions (A was shown first 70 times, B 84 and C 86), so the order effect adds noise and makes judgments order-dependent rather than systematically favouring one condition. For OER, which requires a comparison with the baseline, the dependence is especially visible: simplified responses judged after the baseline had a judged OER of 10.0%, against 1.2% when the baseline came later.

After adjusting the GEE for position, the DAR effect remains large and significant (B vs A OR $= 30.97$, 95% CI $[7.65, 125.48]$, $p < 0.001$; C vs A OR $= 39.81$, 95% CI $[9.82, 161.37]$, $p < 0.001$), and the SMR contrasts remain non-significant (B vs A OR $= 1.47$, 95% CI $[0.89, 2.42]$, $p = 0.134$; C vs A OR $= 1.13$, 95% CI $[0.66, 1.94]$, $p = 0.643$), essentially unchanged from the unadjusted estimates. When only the responses shown first in each group are analysed, SMR was not higher in the simplified conditions (15.7% in A, 7.1% in B, 7.0% in C), while DAR remained elevated (1.4% in A, 13.1% in B, 16.3% in C). These analyses indicate that the DAR finding is robust to the order effect, whereas SMR and especially OER judgments depend on presentation order and should be treated as less reliable.

Table 16. Position effects: judge decisions by position (Panel A), GEE models adjusted for position (Panel B), and rates for responses shown first in each group (Panel C).

Panel A: Judge decisions by position					
Position	*n*	SMR (%)	DAR (%)	OER (%)	Accuracy
1	240	9.6	10.8	0.6	4.50
2	240	12.5	15.8	4.7	4.38
3	240	21.2	19.2	11.2	4.13
Panel B: GEE models adjusted for position					
Outcome	Contrast	OR	95% CI	*p*	
SM	B vs A	1.47	0.89–2.42	0.134	
SM	C vs A	1.13	0.66–1.94	0.643	
DA	B vs A	30.97	7.65–125.48	< 0.001	
DA	C vs A	39.81	9.82–161.37	< 0.001	
Panel C: Responses shown first in each group					
Condition	*n*	SMR (%)	DAR (%)	OER (%)	Accuracy
A (standard)	70	15.7	1.4	–	4.51
B (simplified)	84	7.1	13.1	1.2	4.48
C (highly simplified)	86	7.0	16.3	0.0	4.51

4.11. Summary of Findings

The key findings of the AccessTrap-CS benchmark can be summarised as follows, mapped to the research questions.

- **RQ1 (SMR).** The data do not support a statistically significant increase in judged strategic misconception rate under simplification. H1 is not supported. Model identity is a much stronger predictor of SMR than condition ($F(6, 699) = 5.76$, $p < 0.001$ for model vs $p = 0.326$ for condition). The GLMM with crossed random effects confirms this (OR = 1.50 and 1.13 for B and C vs. A, both non-significant), and its random-effects decomposition shows that the between-model standard deviation (0.95) exceeds the between-concept standard deviation (0.65). The human reference standard on the validation subsample shows the opposite pattern, with SMR decreasing from A to C, which further undermines any claim about the direction of the condition effect on SMR.
- **RQ2 (DAR).** The data strongly support the hypothesis that simplification increases the unconditional dangerous analogy rate. DAR rises from 0.8% in A to 20.4% in B and 24.6% in C, with GEE odds ratios of about 31 and 39 and GLMM odds ratios of 36 and 47 for the simplified conditions relative to baseline. H2 is supported.
- **RQ3 (frequency vs quality).** The unconditional DAR increase is driven primarily by analogy frequency, not by per-analogy danger. On the validation subsample, the standard condition contained no analogies, while both simplified conditions contained an analogy in every response; the conditional DAR was 16.7% in B and 10.0% in C, lower than the unconditional rate and with B above C. H3a is supported; H3b is supported in the sense that conditional DAR is non-negligible, but it does not increase with compression.
- **RQ4 (OER).** Oversimplification errors are measurable. The judge-based rate is greater than zero in B and C, but the expected increase from B to C does not appear (GLMM OR = 0.60 for C vs. B, non-significant), and the derived measure partly reflects sampling noise. H4 is partially supported: the existence claim is supported, the monotonicity claim is not.

- **RQ5 (moderators).** Model is a strong source of variation in all error rates and accuracy. Subfield shows descriptive variation but the effects do not survive multiple-comparison correction. The model × condition interaction is not significant.

The judge validation analysis adds a crucial caveat: the LLM judge has very low recall relative to human reviewers, so the absolute rates reported here are likely underestimates, and the judge disagrees with the humans about the shape of the condition effect on SMR. The relative comparisons across conditions are more robust for DAR, but the position analysis shows that SMR and OER judgments depend on presentation order. The DAR effect is the one finding that withstands all the sensitivity checks we ran, including the GLMM with crossed random effects.

5. Discussion

This section interprets the findings of the AccessTrap-CS benchmark in the context of the accessibility trade-off hypothesis, existing literature on misconceptions in computer science education, and the design of AI tutoring systems. Where the GEE and GLMM estimates differ in magnitude, we report both and interpret the more conservative.

5.1. Testing the Accessibility Trade-off: Mixed Evidence

The results offer a nuanced verdict on the accessibility trade-off. The hypothesis that simplification increases strategic misconceptions (H1) is not supported: SMR is numerically higher in condition B but not significantly so, and condition C is not higher than condition B. The hypothesis that simplification increases dangerous analogies (H2) is strongly supported, with large odds ratios for both simplified conditions relative to baseline, in both the GEE (OR = 30.53 and 38.79) and the GLMM (OR = 36.03 and 47.34). The hypothesis that simplification creates a new class of errors (H4) is partially supported: oversimplification errors are measurable, but they do not increase monotonically with simplification intensity. The decomposition of the DAR gap into frequency and per-analogy danger (RQ3) shows that the accessibility trade-off operates mainly through the first channel.

This pattern refines the accessibility trade-off. The danger of simplification is not a general increase in all types of errors; it is a near-total conversion of explanations into analogy-based explanations, coupled with a non-negligible per-analogy failure rate. When a model is asked to simplify for a student with reading difficulties, it reaches for concrete analogies in essentially every response, and roughly one in six of those analogies is dangerous. The model does not become less accurate in a global sense—accuracy drops moderately, and SMR does not rise significantly—but it shifts the kind of explanation it provides, from technical description to metaphor, and metaphor is where the errors concentrate. The accessibility trade-off, in its empirically supported form, is a frequency-driven analogy problem.

This interpretation is consistent with the literature on analogies in science education. Gentner [50] showed that good analogies transfer relational structure, not surface attributes. The analogies that LLMs produce under simplification pressure tend to transfer surface attributes (a semaphore signals go/stop; a deadlock is a stuck doorway) while dropping the relational structure (atomicity, counting semantics; resource holding, circular wait). Stone et al. [23] documented this pattern for GPT-4 under standard prompting; our results show that accessibility prompts reliably trigger it.

The finding that condition C does not have higher SMR than condition B admits several interpretations. One is that the five-sentence limit leaves so little room that models sometimes avoid developing the analogy into a full misconception. Another is that the judge's low recall makes the SMR estimates noisy, and the difference between 17.1% and 13.8% is not meaningful. A third is that the relationship between simplification intensity

and error rate is genuinely non-monotonic. Because neither the B–A nor the C–B difference is significant, we do not interpret the numerical ordering of B and C; at most, the data give no evidence of a dose-response relationship between simplification intensity and misconceptions.

A generalised linear mixed model with crossed random effects for concept and model (Table **??**) confirms the GEE results: the DAR effect remains large and significant (OR = 36.03 and 47.34 for B vs A and C vs A, both $p < 0.001$), while the SMR contrasts remain non-significant (OR = 1.50 and 1.13, $p = 0.129$ and $p = 0.663$). The GLMM random-effects decomposition is itself informative: for SMR and DAR, the between-model standard deviation ($\sigma_m = 0.95$ and 0.81) exceeds the between-concept standard deviation ($\sigma_c = 0.65$ and 0.58), whereas for OER the pattern is reversed ($\sigma_c = 0.62$, $\sigma_m = 0.32$). This is a formal confirmation of the descriptive pattern in Table 12 and Table 9: strategic misconceptions and dangerous analogies are primarily a property of the model, whereas oversimplification errors are primarily a property of the concept. Practically, this means that switching to a safer model is a more effective mitigation for dangerous analogies than rewriting the prompt, while for oversimplification errors, targeted intervention on the specific concepts that are most vulnerable is more promising than a model-wide fix.

The model comparison results add a further dimension. The between-model variation in SMR (0.0% to 26.5%) dwarfs the between-condition variation (12.5% to 17.1%). This means that the accessibility trade-off, while real for analogies, is not the dominant source of error in LLM-generated CS explanations. By the judge's classification, a student using gemma4:31b in the simplified condition would encounter fewer misconceptions than a student using mistral:7b in the standard condition. The practical implication is that model selection is at least as important as prompt design for safe assistive use.

The judge validation results are a serious limitation. In the validation subsample, the human reviewers flagged 64.4% of responses as containing a strategic misconception, while the judge flagged only 5.6%, more than a tenfold difference. If the human threshold is the relevant one, the true SMR in this corpus is likely several times higher than the judged rate. The judge's precision is moderate, so the responses it does flag are often genuine errors, but its recall is so low that the rates in Table 5 are probably underestimates. The relative comparisons across conditions may still be valid if the judge's conservatism is constant, but we could not verify this, and the position analysis (Table 16) shows that the judge's SMR and OER decisions depend on presentation order: it flagged more errors the later a response appeared. Adjusting for position did not change the SMR contrasts, which remained non-significant. The DAR effect, by contrast, survives position adjustment with a very large odds ratio, and it is confirmed by the GLMM, which strengthens confidence in that finding.

The gap between the judge and the human annotators is not only a matter of overall magnitude. On the human side, the SM rate is highest in condition A and decreases toward condition C; the judge instead reports a non-significant rise in B and no change in C. On the human side, the DA rate peaks sharply in B and stays high in C; the judge reports the same shape but roughly a third of the magnitude. On the human side, the OER rate is above 45% in both simplified conditions; the judge reports between 4% and 7%. The judge, in other words, underestimates the prevalence of every error category and additionally disagrees with the annotators about the shape of the condition effect on SMR. This is a strong argument for treating the judged condition-level rates as under-calibrated indicators of relative risk rather than as estimates of true prevalence.

5.2. Qualitative Analysis of Judge–Human Disagreement

The validation analysis is not an abstract statistical caveat; it reflects concrete cases where the judge and the human experts disagreed. Consider the following example from the adjudication file. For the response on "Garbage Collection (Mark-and-Sweep vs Reference Counting)" generated by gemma4:31b under condition C (highly simplified), the LLM judge classified the response as containing no dangerous analogy (DA = 0). One of the human annotators disagreed, arguing that the response's central analogy—comparing the mark-and-sweep process to "holding hands"—was dangerous because it suggests an undirected connection between objects, whereas the actual mechanism relies on *directed reachability from root references*. The other annotator initially read the analogy as merely incomplete rather than misleading. After discussion in the adjudication meeting, the two annotators agreed that the analogy was dangerous (DA = 1), because the false inference it invites (undirected connectivity) concerns a key property of the algorithm (directed reachability). This is the kind of subtle, plausibility-driven error that the LLM judge missed in most cases, and it illustrates why the judged DAR of 1.9% for gemma4:31b (pooled over conditions) is probably an understatement. Whether the condition effect on DAR is also confirmed by the human annotations requires a condition-level analysis of the human labels, which we leave to future work with a larger annotated sample.

A second pattern from the adjudication file is worth noting. All of the disagreements between the two human annotators occurred in conditions B and C—none occurred in condition A. For responses generated under the standard prompt, the annotators agreed on every item; the simplification prompts introduced ambiguity that even trained experts initially resolved differently. This is consistent with the idea that simplification does not merely introduce errors, it introduces *interpretive ambiguity*, in which the boundary between a helpful simplification and a misleading one becomes contestable. The adjudication meeting resolved these ambiguities by returning to the reference definition and asking whether the response would lead a student to a false belief; in six of the eight disagreement cases, the answer was yes. The human reviewers agreed with each other at $\kappa \geq 0.899$, so the gap with the judge is not an artefact of a noisy annotation process, but it may in part reflect a stricter threshold applied by the human experts than the one implemented by the judge.

5.3. Why Vulnerable Learners Are Disproportionately at Risk

The interpretation below is theoretical and is offered as a framework for reading the results; our experiment does not measure learner cognition (see Section 7). The first reason is asymmetry of verification. A simplified explanation is not merely less detailed; it removes the cues a reader would use to notice that something is missing. Errors of omission are invisible to the person who does not already possess the omitted content, and novices lack the schemas that would flag the gap [42]. The second is load. If decoding or attention regulation already absorbs a large share of working-memory resources [44–46], then the additional operation of checking a claim against prior knowledge competes with comprehension itself, and the reader is likely to fall back on the cheap intuitive judgement [47]. The third is that simplification makes the wrong explanation more persuasive, not less: fluent statements gain a truth premium [17,18], and a short self-contained account produces a feeling of understanding that is not tested [19]. The fourth is the role of analogy. A vivid analogy is memorable by design, so a flawed one is also memorably flawed; it can also function as a seductive detail that captures attention at the expense of the technical core [49], and a single analogy without a stated limit is the textbook recipe for analogy-induced misconception [51,52]. Finally, the deployment context removes external checks: the tutor is private, and there is often no teacher, peer, or textbook running in parallel.

The decomposition of the DAR gap into frequency and per-analogy danger sharpens this argument. What simplification does, according to our data, is not primarily to increase the danger of each analogy. It is to flood the explanation with analogies: in the validation subsample, essentially every simplified response contained at least one. Even if each analogy had the same danger rate as in a standard explanation, the sheer volume of analogies would translate into a correspondingly higher absolute exposure to dangerous analogies. For a reader who is already investing most of their working memory in decoding, the cost of evaluating each analogy against prior knowledge is high, and the number of analogies that need to be evaluated is high as well. The two pressures compound.

It is important to state what this argument does not claim. It does not claim that learners with dyslexia or ADHD are less capable of critical thought. The vulnerability is situational and produced jointly by the learner's load and the properties of the material; the same reasoning predicts risk for any novice reading under time pressure or fatigue. It also does not claim that simplification should be abandoned. The literature on dual-process reasoning is a framework for organising these considerations rather than a validated model of the specific mechanism, and only a study with learners can establish how strongly each factor contributes.

5.4. The Systematic Nature of Oversimplification Errors

The results provide some empirical grounding for the OER construct, with the caveats set out in Section 4. The judge-based OER rate of 6.7% in condition B and 4.2% in condition C indicates that a measurable share of simplified responses contain an error that the same model did not produce for the same concept under standard prompting, and the derived measure (14.2% and 11.7%) points in the same direction, although part of it reflects sampling noise because the opposite transition also occurs. The GLMM result (OR = 0.60 for C vs. B, non-significant) confirms that OER does not grow with simplification intensity. The concept-level analysis suggests where such errors concentrate: the largest SMR increases were for Semaphores and synchronization (+35.7 percentage points), Normalization and CAP theorem (+28.6 each), Dynamic Programming vs Greedy and Transaction Isolation Levels (+25.0 each), and ACID properties and TLS handshake (+21.4 each). These are concepts defined by necessary conditions, invariants, or boundary cases—the properties that analogy-based simplification tends to drop. This observation is post hoc and exploratory: each estimate rests on at most seven concept-model pairs. It is consistent with the GLMM random-effects decomposition for OER ($\sigma_c = 0.62 > \sigma_m = 0.32$), which shows that OER is more a property of the concept than of the model.

The theoretical argument for why oversimplification errors should be systematic runs as follows. A CS concept is defined by a small set of properties, several of which are relational or conditional (invariants, necessary conditions, boundary cases). An analogy typically maps some attributes of a familiar situation onto the target but not its relational structure [50]. When a prompt fixes a strong compression budget (short sentences, five sentences in total, a single analogy), the model has to choose which properties to retain, and the ones that are easiest to render in everyday language are the ones that survive. Necessary conditions (circular wait, atomic) and quantifiers (only if, at least) are the first to be dropped, because they are the least concrete. The resulting error is thus predictable from the structure of the concept and the size of the compression budget, which is why we regard it as systematic.

We distinguish five illustrative subtypes of such errors, which we did not code systematically in this study: (i) omission of a necessary condition; (ii) promotion of an analogy from illustration to definition; (iii) over-generalisation of a property that holds only in a spe-

cial case; (iv) conflation with a neighbouring concept; and (v) replacement of a mechanism by an outcome. Appendix E gives examples.

The long-term consequences are amplified by the hierarchical nature of CS knowledge: later topics (scheduling, synchronisation, transactions, congestion control) build on earlier ones [24], and, once formed, an intuitively appealing wrong model resists later instruction [12,13]. An oversimplification error at the first exposure may therefore become a persistent obstacle that must later be unlearned. The concepts with the highest Δ SMR are those that anchor later courses: semaphores in concurrency, normalization in database design, CAP theorem in distributed systems. An error in these foundations is not an isolated gap; it is a crack in the load-bearing structure of the student's mental model.

5.5. Implications for AI Safety in Education

Three implications follow from the design of the study, independently of the specific numerical results. First, accuracy on a neutral prompt is an insufficient safety criterion for an educational assistant. Models must be evaluated conditioned on the audience and instruction they will actually receive, and benchmark suites should include persona- and format-conditioned variants [29]. The dramatic increase in DAR under accessibility prompts shows that a model can appear safe on standard prompts while producing dangerous analogies when asked to simplify for a student with a disability. Second, standard quality metrics (readability, fluency, coherence) can be optimised in ways that mask semantic degradation: a response can improve on every conventional measure of accessibility while getting worse on conceptual fidelity [31,32]. Condition B is the most readable condition by Flesch Reading Ease, and it also shows a markedly higher DAR than the baseline and the numerically highest SMR (although the SMR difference is not significant). Third, the frequency-versus-quality decomposition shows that the safety-relevant property is not only the per-response error rate but also the analogy density: a model that produces one dangerous analogy per five responses and a model that produces five dangerous analogies per five responses can have the same conditional danger rate but very different effects on a learner.

Mitigation can be pursued at two levels. At the model or prompt level, simplification instructions can be rewritten so that boundary conditions are preserved (e.g., requiring an explicit statement of where each analogy fails, in line with the teaching-with-analogies model [52]), and generate-then-verify pipelines can compare a simplified explanation with a ground-truth checklist. The finding that DAR is the most sensitive metric, and that it is driven primarily by analogy frequency, suggests that analogy-specific interventions—requiring the model to state the limits of each analogy, or to provide multiple analogies with complementary strengths, or to alternate analogies with technical statements—are the most promising prompt-level fix. The GLMM random-effects decomposition suggests an additional mitigation strategy: because the between-model standard deviation dominates for DAR, switching from a higher-risk model to a lower-risk one (e.g., from mistral:7b to gemma4:31b in our corpus) may be more effective than prompt engineering alone. At the interface level, uncertainty and analogy status can be surfaced to the learner, which is the subject of Section 6. Neither level is evaluated in this paper; we present them as hypotheses that follow from the mechanism above.

6. Design Principles for Accessible AI Tutoring Interfaces

The findings of this study have direct implications for the design of AI tutoring interfaces, particularly for students with learning difficulties. Rather than relying exclusively on text-based disclaimers (e.g., AI can make mistakes), which vulnerable learners may overlook or ignore, we propose three accessibility-first design principles that leverage

visual signalling to communicate model uncertainty without increasing cognitive load. The principles are derived from the mechanism analysis in Section 5 and from established work on accessible design, and they are proposals: their effectiveness for the target population has not been tested here.

The principles draw on three bodies of work. Universal Design for Learning and ability-based design argue that systems should adapt to the learner's abilities and provide multiple means of representation instead of adding remedial layers [66,67]. Research on visual perception shows that certain features (colour, shape, orientation, motion) are detected pre-attentively, before effortful reading begins [68]. Research on uncertainty communication shows that visualisation of uncertainty can be effective but requires careful evaluation, and that how a model expresses uncertainty changes users' reliance and trust [69].

6.1. Principle 1: Visual Uncertainty Signalling

Standard text-based disclaimers are cognitively opaque: they occupy prominent screen real estate but are easily dismissed as generic boilerplate. For students with reading difficulties or attention challenges, such disclaimers represent additional text to decode rather than actionable information. We propose replacing or augmenting text disclaimers with visual uncertainty indicators that can be processed pre-attentively [68].

- **Confidence colour coding.** Responses generated with low model confidence are displayed with a subtle amber background. High-confidence responses use a neutral or green background. Colour must not be the only channel (see Principle 3); it is paired with the icon below.
- **Uncertainty icons.** A small, standardised icon (e.g., a dotted outline or question mark) appears adjacent to responses containing high uncertainty. The icon persists even when the user scrolls, providing continuous awareness of reliability.
- **Progressive disclosure of uncertainty details.** Tapping or hovering over the uncertainty indicator reveals a plain-language explanation. This design places uncertainty information at the point of need without overwhelming the primary interface.

Caveat on the signal source. The interface can only convey what the system can estimate. Verbalised model confidence is often poorly calibrated, and better indicators require additional computation such as semantic entropy or consistency across samples. A miscalibrated indicator is arguably worse than none, because it produces unwarranted trust. The judge validation results in this study (Table 15) are a cautionary example: an LLM evaluator with low recall would produce a visual uncertainty signal that flags only a small fraction of the responses a human expert would flag, creating a false sense of safety.

6.2. Principle 2: Analogy Status Tagging

Dangerous analogies are the most robust finding of this study, yet analogies remain a critical accessibility tool for making abstract concepts concrete. Rather than eliminating analogies, interfaces should tag them explicitly, distinguishing between pedagogical analogies (acknowledged as simplifications) and false equivalences (presented as literal descriptions). This operationalises the central recommendation of the teaching-with-analogies model, that the learner be told where the analogy holds and where it breaks [52]. Because the frequency-versus-quality decomposition shows that the danger under accessibility prompts is driven primarily by the sheer volume of analogies, tagging interventions are a natural match to the empirical pattern.

- **Analogy badges.** Each analogy in the generated response is identified and tagged with a small badge reading "Analogy" or "Simplification." The badge serves as a

metacognitive prompt, reminding the learner that the statement is a comparison, not a definition.

- **Toggle between analogy and technical description.** For responses containing analogies, users can toggle to a technical version that replaces the analogy with precise terminology.
- **Analogy boundary highlighting.** The specific sentence or clause containing the analogy is highlighted with a dashed border, making visually explicit where the model has shifted from technical description to figurative comparison.
- **Limits-of-analogy statement.** Each analogy is accompanied, on demand, by a one-line statement of the main property it does not share with the target (e.g., a semaphore also has a counter and cannot be interrupted midway), and, where useful, a second analogy that highlights a complementary aspect [51].

A practical route to badges and boundaries is to require the model to return structured output in which analogy spans are marked; the reliability of such self-tagging is an empirical question. The DAR results suggest that this principle should be applied regardless of the simplification condition: even standard explanations occasionally contain dangerous analogies, and the rate rises sharply as soon as the prompt requests them.

6.3. Principle 3: Accessibility-First Without Text-Based Disclaimers

Current AI interfaces often default to text-based safety disclaimers that are inaccessible to the very populations they aim to protect. Students with dyslexia may skip long disclaimers entirely; students with ADHD may lose focus before reaching the disclaimer. The proposed principles invert this logic: safety information should be integrated into the primary visual channel rather than relegated to peripheral text.

- **No mandatory text disclaimers.** Text-only disclaimers are neither necessary nor sufficient for accessibility. When they must be included (e.g., for regulatory compliance), they should be collapsible with a visual indicator of their collapsed state.
- **Visual-first, not visual-only, communication.** Critical safety information should be communicable through colour, shape, and iconography, with text as a secondary channel for users who opt to explore details. Every visual cue must also have a programmatic text alternative, because many learners with dyslexia use text-to-speech or screen readers, and because colour must not be the sole means of conveying information [70].
- **Restrained motion.** Animation may be offered to attract attention to a signal, but it is optional and off by default under a reduced-motion preference, since moving elements can themselves be distracting for learners with attention difficulties.
- **Customisable visual themes.** Users should be able to select visual themes optimised for their specific needs (e.g., high-contrast mode for dyslexia, reduced motion for attention difficulties) without losing uncertainty signalling [66,67].

These principles shift responsibility from the learner (who must remember to be sceptical) to the interface (which designs scepticism into the visual language). For students with reduced spare capacity for parallel fact-verification, this shift is plausibly not merely a convenience but a prerequisite for safe AI-assisted learning.

6.4. Tensions and Evaluation Roadmap

Two tensions should be acknowledged. Interventions that force the learner to engage (cognitive forcing) reduce over-reliance but increase effort, and are less liked [48]; for the target population that additional effort has a cost, so the principles favour passive, glanceable signals over forcing functions. In addition, adding indicators adds interface elements and therefore load, so the signals must be few, consistent, and learnable [44]. Evaluating

the principles requires a study with learners with and without dyslexia and ADHD, comparing interfaces with and without the signals on measurable outcomes: acceptance of a deliberately flawed explanation, later unassisted performance (following the logic of [11]), perceived workload, and usability. We propose this as future work.

7. Limitations

This study has several limitations that should be considered when interpreting the findings.

7.1. Generalisation to Other Populations

The evaluation in this study was performed by an LLM judge and a small human subsample, not by students with learning difficulties. While this design provides scalable and reproducible evaluation with a validation anchor, it does not measure actual student outcomes. Whether strategic misconceptions produce measurable learning deficits in dyslexic or ADHD students, and whether the plausibility of a flawed explanation predicts actual student belief change, remain open empirical questions. The mechanisms in Section 5 are theoretical. Moreover, the dyslexia and ADHD conditions are prompt personas: they simulate an interaction style and cannot represent the heterogeneity of real learners. The prompts encode a specific assumption—that these learners benefit from simplification and analogy—which is itself a design choice rather than an established requirement of accessibility. Future work should replicate this study with target population samples using learning outcome measures, and should test alternative prompt strategies that do not simply replace precision with analogy.

7.2. Generalisation to Other Domains

The benchmark is restricted to computer science education, and to English-language prompts. Other STEM domains (e.g., physics, mathematics) and non-STEM fields (e.g., history, law) may exhibit different simplification dynamics. For example, fields with higher tolerance for interpretive variation may show lower rates of oversimplification errors, while fields with axiomatic knowledge structures may show higher rates. Extending AccessTrap-CS to additional disciplines and languages is a priority for future work.

A natural follow-up question is whether human-authored simplifications for the same concepts exhibit lower dangerous analogy rates. If human teachers, writing simplifications for the same students, produce dangerous analogies at a rate comparable to the LLM rate observed here, then the accessibility trade-off is a general property of simplification for technical content rather than an LLM-specific failure mode. If human-authored simplifications are safer, then the difference identifies a concrete target for model improvement. This comparison was outside the scope of the present study, but it is an important benchmark for future work.

7.3. Generalisation to Other Models

The seven evaluated models, while diverse in scale and family, represent only a fraction of available LLMs, and none exceeds 32B parameters. Open-weight models were chosen deliberately to ensure full reproducibility and to allow other researchers to rerun the entire pipeline locally, without depending on a vendor's changing API or on a paid subscription. This design choice prioritises reproducibility over coverage, and we regard it as a strength of the benchmark rather than a limitation: any result reported here can be independently verified. Proprietary frontier models were not included because of their closed weights and API costs. It remains unknown whether the pattern observed here generalises to these systems, which may incorporate additional safety fine-tuning. Ollama serves quantised weights by default, so behaviour may differ from full-precision deployments.

7.4. Prompt Design and Construct Validity

As explained in Section 3, the conditions are bundles of constraints; the accuracy cue appears only in condition A; and condition B explicitly discourages formal definitions. The results therefore describe the behaviour of models under realistic accessibility-style requests, not the causal effect of any single element such as analogy use or length limit. Ablation prompts (e.g., simplification with an explicit accuracy instruction, or a single-analogy requirement without an audience label) would separate these factors. Each cell also contains a single generation per concept-model-condition, so the estimates reflect between-concept and between-model variability but not within-cell sampling variability.

7.5. Dangerous Analogy Rate: Frequency Versus Quality

The unconditional DAR reported in Section 4 combines two distinct quantities: the frequency with which models produce analogies, and the per-analogy danger of those analogies. The decomposition reported in Section 4.4 shows that, on the human validation subsample, the frequency component saturates: the standard condition contained no analogies, while both simplified conditions contained an analogy in every response. Conditional DAR is therefore estimable only for B and C on this subsample; condition A is undefined, and the confidence intervals on the conditional rates are wide (30 responses each). The manual coding of analogy presence was performed by the first author with the condition label masked, and any disagreements were resolved by re-reading the response; we did not have a second independent coder for analogy presence, so the conditional rates inherit any idiosyncrasy of the single coder's operationalisation. We treat the conditional DAR as an exploratory estimate that constrains the interpretation of the unconditional DAR, not as a definitive measure of per-analogy quality. A follow-up analysis conditional on analogy presence on the full corpus, with multiple independent coders and an ablation in which the baseline also requests an analogy, would fully separate the two channels.

A second independent coder for analogy presence would allow us to report inter-coder agreement on this indicator and to rule out idiosyncrasy in the operationalisation; we treat this as a priority for the replication.

7.6. Causal Interpretation of Oversimplification Errors

The classification of oversimplification errors relies on the assumption that an error present in a simplified response but absent from the baseline response for the same concept and model was caused by the simplification pressure. While this design controls for concept and model variation, it does not control for random variation in model outputs. Because generation was performed with temperature $T = 0.7$, two generations from the same model under identical conditions could produce different outputs, so an error attributed to simplification might have appeared in a different baseline sample; the frequency of the opposite transition (Section 4) shows that this is not a minor concern. Replication with temperature $T = 0.0$, or with several samples per cell, would isolate the simplification effect but would also reduce ecological validity.

7.7. LLM-as-Judge Validation

Qwen3.6:35b was used as the primary evaluator. The agreement analysis (Table 15) shows that the judge has very low recall: it identifies only 5.2% of the strategic misconceptions that human reviewers flag, 19.4% of dangerous analogies, and 5.9% of oversimplification errors. The absolute rates reported in Section 4 are therefore likely substantial underestimates relative to the human standard. High agreement between the two human reviewers shows consistency, not that their threshold is the only defensible one, and the large prevalence gap may partly reflect a stricter human threshold. The human evaluation

was limited to a stratified subsample of 30 concept-model groups (150 ratings per annotator across Parts 1 and 2), not the full set of 720 judged responses. While this subsample was stratified by subfield and by model, it is possible that agreement varies across concepts or conditions in ways that the subsample cannot capture, and we did not analyse the human labels by condition.

We therefore recommend that the judged rates reported in this paper be interpreted as a lower bound on the true error prevalence, not as point estimates. The relative ordering of conditions, rather than the absolute magnitude of the rates, should be the primary object of inference, and even the relative ordering should be read with caution where the judge and the human annotators disagree about the shape of the effect (as they do for SMR). A larger validation study, ideally with three or more annotators per item and full-corpus coverage, is the most important methodological priority for future work.

7.8. Evaluator Discrepancy and Position Bias

Two additional limitations of the automated evaluation pipeline deserve to be stated explicitly, because they bound the conclusions that can be drawn from the judged rates.

First, the evaluator discrepancy is not merely numerical. On the validation subsample, the judge and the annotators disagree not only about the magnitude of the error rates but also about their shape across conditions: the human annotators report an SM rate that decreases from A to C, whereas the judge reports a non-significant rise in B and no change in C. This means that the null result for H1 on the judged data is weaker than it would be if the judge had agreed with the human pattern. It also means that the SMR comparisons across conditions should not be read as evidence that simplification has no effect on misconceptions; the human data suggest the opposite trend, and the judge's failure to reproduce it may be due to its low recall for precisely the subtle, plausibility-driven errors that the human reviewers identify.

Second, the position-bias analysis (Table 16) shows a pronounced order effect: the judge flags roughly twice as many errors at presentation position 3 as at position 1, and the effect is especially strong for OER, which depends on whether the baseline was seen before the simplified response. Adjusting for position does not change the DAR or SMR contrasts, so the main findings are not artefacts of the order effect, but the SMR and OER comparisons remain order-dependent and should be interpreted with corresponding caution.

To mitigate these limitations in future work, we advocate for: (i) utilising human-in-the-loop expert validation for high-stakes pedagogical error detection, especially for the construct classes whose rate the judge underestimates most severely (SM and OER); (ii) employing multi-judge consensus architectures with randomised candidate position swapping, so that no condition is systematically favoured by the order of presentation; and (iii) fine-tuning specialised evaluator models on domain-specific pedagogical error taxonomies rather than relying on general-purpose zero-shot LLM evaluators, whose sensitivity to subtle conceptual errors is by construction limited by their training objective.

7.9. Parsing Failures and Output Length

DeepSeek-R1 exhibited parsing issues due to its unique output format, which wraps reasoning traces within `<think>` tags. In some instances, these tags interfered with programmatic segmentation, resulting in unrecoverable text segments, and 15 responses from five concept-model groups were lost for the judged analyses. Because reasoning tokens count toward the generation limit, a cap of 512 tokens may also truncate the visible answer of reasoning models. This limitation is specific to reasoning-style output and does not affect the other models, but it highlights a broader challenge: as models evolve, they may adopt

output formats that break standard parsing assumptions. Future versions of the benchmark should implement model-specific parsers and length budgets.

7.10. Statistical Model Specification

The GEE model reported as the primary analysis clusters at the level of the concept-model pair, but does not separately estimate the variance attributable to concept and to model. We addressed this by fitting a generalised linear mixed model with crossed random effects for concept and model (Section 3.9.3), and we report both estimates side by side. The GLMM and GEE agree qualitatively on all three outcomes: SMR contrasts remain non-significant, DAR contrasts remain large and highly significant, and OER does not increase from B to C. The GLMM estimates for DAR are somewhat larger than the GEE estimates (OR = 36.03 and 47.34 versus 30.53 and 38.79), which means that the GEE result, if anything, understates the effect. The GLMM also produces a between-model and between-concept variance decomposition that the GEE cannot, and this decomposition is itself substantively informative: model dominates for SM and DA, concept dominates for OER. A limitation that remains is that the GLMM is fitted with the Laplace approximation, which can be less accurate than adaptive Gauss-Hermite quadrature for small cluster counts; however, with 245 clusters and 720 observations, the Laplace approximation is generally adequate. A further limitation is that the GLMM assumes normally distributed random intercepts on the logit scale; alternative distributions (e.g., *t*-distributed) were not tested.

8. Conclusions

This paper introduced the accessibility trade-off: the hypothesis that LLM simplification for students with learning difficulties, while well-intentioned, shifts the risk profile of generated explanations from technically precise descriptions to memorable but misleading analogies. We presented AccessTrap-CS, a benchmark corpus of 735 responses generated by seven open-weight LLMs across 35 core computer science concepts under three simplification conditions, together with an error taxonomy comprising strategic misconceptions, dangerous analogies, and oversimplification errors, and an LLM-as-judge evaluation with Qwen3.6:35b as the evaluator (720 responses judged), validated against a stratified human-annotated subsample of 90 responses that was also manually coded for analogy presence. Statistical analysis combined descriptive tests and GEE in Python with a crossed random-effects GLMM in R.

The empirical findings offer a nuanced picture. The strongest effect is on dangerous analogies: judged DAR rises from 0.8% in the standard condition to 20.4% in the simplified condition and 24.6% in the highly simplified condition, with GEE odds ratios of about 31 and 39 and GLMM odds ratios of 36.03 and 47.34. This effect survives adjustment for the judge's order effect, a restriction to first-shown responses, and the switch to a crossed random-effects specification, although the odds ratios are imprecise because the baseline contains only two events. The frequency-versus-quality decomposition on the validation subsample shows that this increase is driven mainly by analogy frequency: the standard condition contained no analogies, while both simplified conditions contained an analogy in every response. Conditional DAR, computed only over responses containing an analogy, is 16.7% in condition B and 10.0% in condition C, lower than the unconditional rate and with B above C. Judged accuracy also falls significantly under simplification (4.67 to 4.15 and 4.18). The strategic misconception rate shows a non-significant trend toward higher values in the simplified condition (17.1% vs 12.5%) but does not increase monotonically with simplification intensity, and the human annotators on the validation subsample report a decrease in SM from A to C, which further undermines any claim about the direction of the condition effect on SMR. The judge-based oversimplification error rate is above

zero in both simplified conditions (6.7% and 4.2%; 14.2% and 11.7% by a derived measure that partly reflects sampling noise), but the expected increase from moderate to extreme simplification does not appear (GLMM OR = 0.60 for C vs. B, non-significant). Model identity is a stronger predictor of error rates than condition: the between-model variation in SMR (0.0% to 26.5%) is much larger than the between-condition variation (12.5% to 17.1%), and the GLMM random-effects decomposition formalises this (between-model SD of 0.95 and 0.81 for SM and DA, versus between-concept SD of 0.65 and 0.58). For OER, the pattern is reversed (between-concept SD of 0.62 versus between-model SD of 0.32), which suggests that oversimplification errors are more a property of the concept than of the model.

The judge validation analysis is a critical caveat. In the validation subsample, human reviewers flagged 64.4% of responses as containing a strategic misconception, while the LLM judge flagged only 5.6%. The absolute rates reported in this paper are therefore likely underestimates relative to the human standard, and the null result for SMR cannot be interpreted as evidence of no effect. The relative comparisons across conditions are more robust, but the judge's SMR and OER decisions depend on presentation order. The DAR effect is the finding that withstands all the sensitivity checks we ran, including the crossed random-effects GLMM.

Beyond empirical findings, this paper contributes three accessibility-first design principles for AI tutoring interfaces: visual uncertainty signalling, analogy status tagging, and visual-first communication that does not rely on text-based disclaimers. These principles shift responsibility for scepticism from the learner to the interface, a design move that is plausibly important for students with reduced spare capacity for parallel fact-verification. The frequency-versus-quality decomposition of DAR suggests that analogy tagging is the most immediately actionable principle, since the dominant risk is the sheer volume of analogies produced under accessibility prompts rather than a change in their per-analogy quality. The GLMM random-effects decomposition suggests a complementary mitigation strategy: because the between-model variance dominates for DAR, switching to a lower-risk model is an effective lever that does not depend on prompt engineering.

Future work should extend AccessTrap-CS to additional domains and languages, add ablation prompts and multiple samples per cell, and replicate the conditional DAR analysis on the full corpus with multiple independent coders. Replicating the study with target population samples using learning outcome measures, and implementing and evaluating the proposed design principles in tutoring systems in real-world settings, are the most important next steps. A priority is to improve the judge: the current LLM-as-judge approach has low recall, and a hybrid human–LLM evaluation or a calibrated judge with higher sensitivity would provide more reliable absolute rates. As LLMs become increasingly integrated into educational practice, the ability to detect, measure, and mitigate strategic misconceptions will be essential for ensuring that accessibility tools do not inadvertently become vectors of conceptual harm. The accessibility trade-off, in the form supported by our data, concentrates in analogies, operates mainly through their frequency, and varies by model; knowing where the risk is concentrated is the first step toward designing it out.

Author Contributions: Conceptualization, M.M., S.B.Š., and I.L.; methodology, M.M., S.B.Š. and D.E.; software, M.M.; validation, I.L., S.B.Š. and D.E.; formal analysis, M.M.; investigation, M.M.; resources, M.M., I.L. and D.E.; data curation, M.M.; writing—original draft preparation, M.M.; writing—review and editing, S.B.Š., I.L. and D.E.; visualization, M.M.; supervision, S.B.Š. and I.L.; project administration, S.B.Š., I.L. and D.E.; funding acquisition, S.B.Š. and I.L.;
All authors have read and agreed to the published version of the manuscript.

Funding: This work was (partially) supported by the EC Digital Europe Programme EDIH Adria 2.0 (101256325); SPIN projects IP.1.1.03.0120, IP.1.1.03.0158 and IP.1.1.03.0039; and NextGenerationEU University grants: uniri-iz-25-6, uniri-iz-25-220, IIP_010144, IIP_010136

Data Availability Statement: The AccessTrap-CS corpus and evaluation scripts are available at [repository URL to be added upon acceptance].

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

LLM	Large Language Model
CS	Computer Science
SMR	Strategic Misconception Rate
DAR	Dangerous Analogy Rate
OER	Oversimplification Error Rate
ADHD	Attention Deficit Hyperactivity Disorder
ANOVA	Analysis of Variance
HPC	High Performance Computing
GEE	Generalized Estimating Equations
GLMM	Generalized Linear Mixed Model
FRE	Flesch Reading Ease
FKGL	Flesch-Kincaid Grade Level

Appendix A CS Concepts and Subfields

Table A1 lists the 35 concepts in AccessTrap-CS, grouped by subfield. Table A2 identifies the concepts with the largest mean increase in SMR from condition A to the average of conditions B and C.

Table A1. The 35 CS concepts in AccessTrap-CS, grouped by subfield.

ID	Concept
Algorithms & Data Structures	
ALG_01	QuickSort (average vs worst-case complexity)
ALG_02	Dijkstra's algorithm (negative edge weights limitation)
ALG_03	Dynamic Programming vs Greedy Approach
ALG_04	Hash tables (collisions and load factor)
ALG_05	Binary search tree balancing (AVL vs Red-Black)
ALG_06	Breadth-First Search (BFS) and Depth-First Search (DFS) — memory consumption
ALG_07	Space Complexity and trade-offs
Computer Networks	
NET_01	TCP three-way handshake (SYN, SYN-ACK, ACK)
NET_02	OSI model vs TCP/IP model (differences in layers)
NET_03	DNS resolution (Recursive vs Iterative)
NET_04	IP addressing and subnets (CIDR mask)
NET_05	HTTP methods: Idempotency of GET, PUT, DELETE versus POST
NET_06	Routing: Distance Vector (RIP) vs Link State (OSPF)
NET_07	TLS handshake (RSA vs Diffie-Hellman key exchange)
Databases	
DB_01	ACID properties (especially Isolation vs Consistency)
DB_02	Normalization (1NF, 2NF, 3NF) and its limits
DB_03	SQL JOIN: INNER vs OUTER (LEFT/RIGHT/FULL)
DB_04	Indexes: When a B-tree index does not speed up a query
DB_05	Transactions: Different isolation levels (Read Committed vs Serializable)
DB_06	CAP theorem (Consistency, Availability, Partition tolerance)
DB_07	Denormalization: When and why to use it
Operating Systems	
OS_01	Processes vs threads
OS_02	Context switching
OS_03	CPU scheduling algorithms
OS_04	Deadlock conditions
OS_05	Virtual memory and paging
OS_06	Paging vs Segmentation
OS_07	Semaphores and synchronization
Programming Languages & Paradigms	
PLP_01	Inheritance and Liskov Substitution Principle (LSP)
PLP_02	Interfaces vs Abstract Classes
PLP_03	Closures and lexical scope
PLP_04	Static vs Dynamic typing (Type safety)
PLP_05	Exceptions: Checked vs Unchecked
PLP_06	Garbage Collection (Mark-and-Sweep vs Reference counting)
PLP_07	Concurrency: Threads vs Asynchronous programming (Async/Await)

The distribution of concepts across subfields reflects the balance of the underlying curriculum: each of the five subfields contributes exactly seven concepts, so that subfield-level comparisons are not driven by unequal sample sizes. Concepts were chosen to span the range from formal algorithmic reasoning (e.g., Dijkstra's algorithm, dynamic programming) through language-level semantics (e.g., closures, Liskov substitution) to system-level mechanisms (e.g., deadlock, paging, TCP handshake). This spread is deliberate, because the accessibility trade-off is expected to manifest differently depending on whether the concept's core is a mathematical invariant, a semantic relation, or a procedural mechanism.

Table A2. Concepts with the largest mean increase in SMR from A to mean(B, C). Each value rests on at most seven concept-model pairs and is exploratory.

ID	Concept	Δ SMR (pp)
OS_07	Semaphores and synchronization	+35.7
DB_02	Normalization (1NF, 2NF, 3NF) and its limits	+28.6
DB_06	CAP theorem (Consistency, Availability, Partition tolerance)	+28.6
ALG_03	Dynamic Programming vs Greedy Approach	+25.0
DB_05	Transactions: Different isolation levels	+25.0
DB_01	ACID properties (Isolation vs Consistency)	+21.4
NET_07	TLS handshake (RSA vs Diffie-Hellman)	+21.4
PLP_05	Exceptions: Checked vs Unchecked	+14.3
PLP_02	Interfaces vs Abstract Classes	+14.3
PLP_07	Concurrency: Threads vs Asynchronous programming	+7.1

The pattern in Table A2 is informative in its own right, although it is post hoc. The concepts with the largest increase in SMR under simplification are those whose definitions hinge on necessary conditions, invariants, or boundary cases: semaphores (atomicity, counting), normalization (functional dependencies and their limits), CAP theorem (the impossibility trade-off), dynamic programming versus greedy (optimal substructure and the exchange argument), transaction isolation levels (which anomalies each level permits), ACID (the distinction between isolation and consistency), and the TLS handshake (forward secrecy and the role of each message). These are the properties that the analogy-based simplification process tends to drop, which is consistent with the theoretical account of OER presented in Section 5 and with the GLMM random-effects decomposition for OER (between-concept SD of 0.62 exceeding between-model SD of 0.32).

Appendix B Prompt Templates

The three templates used in the study are reproduced verbatim below. `[CONCEPT]` is replaced by the concept name.

Condition A (standard).

```
Explain the following computer science concept accurately and clearly,
suitable for a first-year computer science student.

Answer in English.

Concept: [CONCEPT]
```

Condition B (simplified, reading difficulties).

```
Explain the following computer science concept to a student who has
reading difficulties. Use simplified language, short sentences, and
concrete analogies instead of code or formal definitions. Avoid
technical jargon where possible.

Answer in English.

Concept: [CONCEPT]
```

Condition C (highly simplified, attention difficulties).

```
Explain the following computer science concept to a student with
significant attention difficulties. Keep the explanation within five
sentences. Use one single, vivid analogy that captures the core idea.
```

```
Make it memorable and engaging.

Answer in English.

Concept: [CONCEPT]
```

The three templates differ only in the instruction line, so that any difference in the resulting responses can be attributed to the instruction rather than to unrelated formatting variation. The prompts are deliberately short, because they are intended to mirror how a student or teacher would actually phrase a request to a general-purpose assistant, not how a carefully engineered benchmark prompt would be constructed. This design choice increases ecological validity at the cost of construct specificity, a trade-off that is discussed in Section 7.

Appendix C Evaluation Codebook

The codebook used for human annotation and for the LLM judge prompt defines each indicator as follows.

Strategic Misconception (SM). Code 1 if the explanation contains a subtle but conceptually significant error that sounds plausible but is fundamentally wrong. The error must be consequential: it would lead a student to construct a mental model that is systematically incorrect. Code 0 if the explanation is accurate or contains only trivial slips (typos, imprecise wording that does not change the concept). Do not code an error that is merely an omission unless the omission changes the meaning of the concept (e.g., omitting atomicity when defining a semaphore changes the concept; omitting a rarely used feature does not).

Dangerous Analogy (DA). Code 1 if the explanation uses an analogy that is intuitive but misrepresents a key technical property of the concept. The analogy must invite an inference that is false for the target. An analogy that is merely incomplete is not dangerous. Examples: semaphore as traffic light (invites inference of simple binary flag, no atomicity); deadlock as two people stuck in a doorway (invites inference that any waiting is a deadlock, omits circular wait); database index as a book index (invites inference that indexes always speed up queries, omits write overhead and selectivity).

Oversimplification Error (OER). Code 1 if the explanation contains an error that would not appear in a standard explanation of the same concept by the same model. This is a comparative judgment across conditions: the annotator must compare the simplified response to the baseline response for the same concept and model. Code 0 if the error also appears in the baseline, or if the simplified response contains no error beyond what the baseline contains.

Accuracy Score. A Likert scale from 1 to 5: 1 = completely wrong; 2 = mostly wrong with some correct elements; 3 = partially correct with significant gaps or errors; 4 = mostly accurate with minor issues; 5 = perfectly accurate.

Analogy Presence (AP). Code 1 if the response contains at least one analogy, metaphor, or simile. An explicit comparison (*like*, *as*, *think of*, *imagine*), a metaphor, or an everyday comparison to a non-technical object or situation counts as an analogy. Formal definitions, purely technical comparisons, and illustrative examples without an external referent do not count. The purpose of this indicator is to allow the unconditional DAR to be decomposed into analogy frequency and conditional per-analogy danger.

The codebook was calibrated by the two annotators on a shared pilot set of 10 concept-model groups prior to the main annotation round. The pilot round surfaced two boundary cases that required clarification. First, an analogy that is factually correct but incomplete should be coded 0 for DA unless the incompleteness invites a specific false inference. Second, an error that is present in the baseline response but only becomes misleading in

the simplified context should be coded 1 for OER only if a reader of the simplified response would form a different (and false) belief than a reader of the baseline. These clarifications were incorporated into the final version of the codebook above.

Appendix D LLM Judge Prompt

The LLM judge was Qwen3.6:35b with temperature $T = 0.0$ and `num_predict` = 1024. The prompt presented the ground truth and the three responses for a single concept-model group in randomised order, and asked the judge to return a JSON object with one entry per response. The instruction block was as follows.

```
You are an expert computer science educator and evaluator.
You will be given a ground-truth definition of a computer
science concept and three explanations of the same concept,
generated by the same model under three different prompt
conditions. The conditions were: (A) a standard explanation
for a first-year computer science student; (B) a simplified
explanation for a student with reading difficulties; (C) a
highly simplified explanation for a student with attention
difficulties. The order in which the three responses are
presented to you has been randomised.

For each response, decide:
1. strategic_misconception (true/false): does the explanation
   contain a subtle but conceptually significant error that
   sounds plausible but is fundamentally wrong?
2. strategic_misconception_explanation (string): if true,
   explain the misconception; if false, write "None".
3. dangerous_analogy (true/false): does the explanation use
   an analogy that is intuitive but misrepresents a key
   technical property of the concept?
4. dangerous_analogy_explanation (string): if true, explain
   why the analogy is dangerous; if false, write "None".
5. oversimplification_error (true/false): does the
   explanation contain an error that would NOT appear in a
   standard explanation of the same concept by the same
   model? (For condition A, code this as false by
   construction.)
6. oversimplification_error_explanation (string): if true,
   explain the error; if false, write "None".
7. accuracy_score (integer 1-5): 1 = completely wrong,
   2 = mostly wrong, 3 = partially correct, 4 = mostly
   accurate with minor issues, 5 = perfectly accurate.

Ground truth: [GROUND_TRUTH]

Response 1 (position 1):
[RESPONSE_TEXT]

Response 2 (position 2):
[RESPONSE_TEXT]
```

```
Response 3 (position 3):
[RESPONSE_TEXT]

Return a single JSON object with three top-level keys
"response_1", "response_2", "response_3", each mapping
to an object with the seven fields above. Do not include
any other text in your output.
```

A 2-second pause was inserted between judge requests to prevent API overload. The judge output was parsed with a JSON parser; unrecoverable parse failures (15 responses from five concept-model groups) were excluded from the judged analyses and reported in Table 3. The manual coding of analogy presence on the human validation subsample did not use the judge prompt; it was performed by the first author following the operational definition in Appendix C.

Appendix E Annotated Examples of Dangerous Analogies

This appendix presents representative examples of dangerous analogies identified in the corpus, with expert explanations of why they are dangerous.

Example 1: Semaphore as traffic light (OS_07, condition C). The model explains a semaphore by comparing it to a traffic light: green means go, red means stop. This analogy captures the binary blocking/unblocking behaviour but omits the counting semantics (a semaphore can allow multiple concurrent accesses) and the atomicity of the wait and signal operations. A student who learns this analogy may believe that a semaphore is a simple boolean flag and may not understand how a counting semaphore differs from a mutex.

Example 2: Deadlock as two people in a doorway (OS_04, condition B). The model describes deadlock as two people trying to pass through a doorway at the same time, each waiting for the other to move. This captures mutual waiting but omits the necessary conditions: resource holding, no preemption, and circular wait. A student may believe that any two processes waiting for each other constitute a deadlock, even if one of them can be preempted or if there is no circular dependency.

Example 3: Database index as book index (DB_04, condition B). The model compares a database index to the index at the back of a book: it helps you find things faster. This captures the read-speed benefit but omits the write overhead, the selectivity requirement, and the fact that an index on a low-selectivity column can be slower than a full table scan. A student may believe that adding an index always speeds up queries.

Example 4: TCP handshake as a greeting (NET_01, condition C). The model describes the TCP three-way handshake as a polite greeting: “Hello?” “Hello, I hear you.” “Great, let’s talk.” This captures the exchange of messages but omits the purpose of the handshake (synchronising sequence numbers, establishing parameters) and the fact that the third message is necessary to confirm that both sides are ready. A student may believe that the handshake is merely a formality and not a functional part of connection establishment.

Example 5: Garbage collection as cleaning up (PLP_06, condition B). The model describes garbage collection as “cleaning up memory that is no longer needed.” This captures the outcome but omits the mechanism (reachability analysis, mark-and-sweep vs reference counting) and the fact that garbage collection can pause execution and that reference counting cannot collect cycles. A student may believe that garbage collection is a background process with no performance implications.

In each case, the analogy satisfies the two defining properties of a dangerous analogy as specified in the codebook: (a) it is intuitive and easy to process, and (b) it invites a specific inference that is false for the target concept. The examples also illustrate the illustrative error subtypes discussed in Section 5 (which were not coded systematically): Example 1 is a case

of omission of a necessary condition; Examples 2 and 4 are cases of over-generalisation of a property that holds only in a special case; Example 3 is a case of promotion of an analogy from illustration to definition; and Example 5 is a case of replacement of a mechanism by an outcome.

References

1. Zdravkova, K.; Krasniqi, V.; Dalipi, F.; Ferati, M. Cutting-edge communication and learning assistive technologies for disabled children: An artificial intelligence perspective. *Frontiers in Artificial Intelligence* **2022**, *5*, 970430. https://doi.org/10.3389/frai.2022.970430.
2. Kortemeyer, G. Could an artificial-intelligence agent pass an introductory physics course? *Physical Review Physics Education Research* **2023**, *19*, 010132. https://doi.org/10.1103/PhysRevPhysEducRes.19.010132.
3. Kasneci, E.; Seßler, K.; Küchemann, S.; Bannert, M.; Dementieva, D.; et al. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences* **2023**, *103*, 102274. https://doi.org/10.1016/j.lindif.2023.102274.
4. Rello, L.; Baeza-Yates, R.; Bott, S.; Saggion, H. Simplify or help? Text simplification strategies for people with dyslexia. In Proceedings of the Proceedings of the 10th International Cross-Disciplinary Conference on Web Accessibility, 2013, pp. 1–10. https://doi.org/10.1145/2461121.2461126.
5. Štajner, S.; Saggion, H. Automatic text simplification for reading assistance: A review of methods, applications, and future directions. *Natural Language Engineering* **2021**, *27*, 507–546. https://doi.org/10.1017/S1351324920000525.
6. Finnie-Ansley, J.; Denny, P.; Becker, B.A.; Luxton-Reilly, A.; Prather, J. The robots are coming: Exploring the implications of OpenAI Codex on introductory programming. In Proceedings of the Proceedings of the 24th Australasian Computing Education Conference, 2022, pp. 10–19. https://doi.org/10.1145/3511861.3511863.
7. Prather, J.; Denny, P.; Leinonen, J.; Becker, B.A.; Albluwi, I.; et al. The robots are here: Navigating the generative AI revolution in computing education. In Proceedings of the Proceedings of the Working Group Reports on Innovation and Technology in Computer Science Education, 2023, pp. 97–149. https://doi.org/10.1145/3623762.3633499.
8. Denny, P.; Prather, J.; Becker, B.A.; Finnie-Ansley, J.; Hellas, A.; et al. Computing education in the era of generative AI. In Proceedings of the Proceedings of the 55th ACM Technical Symposium on Computer Science Education, 2024. https://doi.org/10.1145/3626252.
9. Pardos, Z.A.; Bhandari, S. ChatGPT-generated help produces learning gains equivalent to human tutor-authored help on mathematics skills. *PLOS ONE* **2024**, *19*, e0304013. https://doi.org/10.1371/journal.pone.0304013.
10. Kestin, G.; Miller, K.; Klales, A.; Milbourne, T.; Ponti, G. AI tutoring outperforms in-class active learning: An RCT introducing a novel research-based design in an authentic educational setting. *Scientific Reports* **2025**, *15*, 17458. https://doi.org/10.1038/s41598-025-97652-6.
11. Bastani, H.; Bastani, O.; Sungu, A.; Ge, H.; Kabakcı, Ö.; Mariman, R. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences* **2025**, *122*, e2422633122. https://doi.org/10.1073/pnas.2422633122.
12. Posner, G.J.; Strike, K.A.; Hewson, P.W.; Gertzog, W.A. Accommodation of a scientific conception: Toward a theory of conceptual change. *Science Education* **1982**, *66*, 211–227. https://doi.org/10.1002/sce.3730660207.
13. Chi, M.T.H.; Slotta, J.D.; de Leeuw, N. From things to processes: A theory of conceptual change for learning science concepts. *Learning and Instruction* **1994**, *4*, 27–43. https://doi.org/10.1016/0959-4752(94)90017-5.
14. Vosniadou, S.; Brewer, W.F. Mental models of the earth: A study of conceptual change in childhood. *Cognitive Psychology* **1992**, *24*, 535–585. https://doi.org/10.1016/0010-0285(92)90018-W.
15. Qian, Y.; Lehman, J.D. Students' misconceptions and other difficulties in introductory programming: A literature review. *ACM Transactions on Computing Education* **2017**, *18*, 1–24. https://doi.org/10.1145/3077618.
16. Sorva, J. Visual Program Simulation in Introductory Programming Education. PhD thesis, Aalto University, 2012.
17. Reber, R.; Schwarz, N. Effects of perceptual fluency on judgments of truth. *Consciousness and Cognition* **1999**, *8*, 338–342. https://doi.org/10.1006/ccog.1999.0392.
18. Alter, A.L.; Oppenheimer, D.M. Uniting the tribes of fluency to form a metacognitive nation. *Personality and Social Psychology Review* **2009**, *13*, 219–235. https://doi.org/10.1177/1088868309341564.
19. Rozenblit, L.; Keil, F. The misunderstood limits of folk science: An illusion of explanatory depth. *Cognitive Science* **2002**, *26*, 521–562. https://doi.org/10.1207/s15516709cog2605_1.
20. Snowling, M.J.; Hulme, C.; Nation, K. Defining and understanding dyslexia: past, present and future. *Oxford Review of Education* **2019**, *46*, 501–513. https://doi.org/10.1080/03054985.2020.1765756.
21. Barkley, R.A. *Attention-Deficit Hyperactivity Disorder: A Handbook for Diagnosis and Treatment*, 4th ed.; Guilford Publications, 2015.

22. Martinussen, R.; Hayden, J.; Hogg-Johnson, S.; Tannock, R. A meta-analysis of working memory impairments in children with attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child & Adolescent Psychiatry* **2005**, *44*, 377–384. https://doi.org/10.1097/01.chi.0000153228.72591.73.
23. Stone, M.; Liao, Y.; Szafir, D.; Boyer, K.E. Analogies in the age of LLMs: When explanatory metaphors mislead CS learners. In Proceedings of the Proceedings of the 55th ACM Technical Symposium on Computer Science Education, 2024, pp. 1256–1262. https://doi.org/10.1145/3626253.3635473.
24. Ben-Ari, M. Constructivism in computer science education. *Journal of Computers in Mathematics and Science Teaching* **2001**, *20*, 45–73.
25. Ibrahim, H.; Liu, T.; Chen, A.; Wang, D. The warmth-accuracy trade-off: When making LLMs more empathetic increases factual errors. In Proceedings of the Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency, 2026. Forthcoming.
26. Ouyang, L.; Wu, J.; Jiang, X.; Almeida, D.; Wainwright, C.; et al. Training language models to follow instructions with human feedback. In Proceedings of the Advances in Neural Information Processing Systems, 2022, Vol. 35, pp. 27730–27744. https://doi.org/10.48550/arXiv.2203.02155.
27. Sharma, M.; Tong, M.; Korbak, T.; Duvenaud, D.; Askell, A.; et al. Towards understanding sycophancy in language models. In Proceedings of the Proceedings of the 12th International Conference on Learning Representations, 2024. https://doi.org/10.48550/arXiv.2310.13548.
28. Perez, E.; Ringer, S.; Lukošiūtė, K.; Nguyen, K.; Chen, E.; et al. Discovering language model behaviors with model-written evaluations. In Proceedings of the Findings of the Association for Computational Linguistics: ACL 2023, 2023, pp. 13387–13434. https://doi.org/10.18653/v1/2023.findings-acl.847.
29. Lin, S.; Hilton, J.; Evans, O. TruthfulQA: Measuring how models mimic human falsehoods. In Proceedings of the Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, 2022, pp. 3214–3252. https://doi.org/10.18653/v1/2022.acl-long.229.
30. Gadiraju, V.; Kane, S.; Dev, S.; Taylor, A.; Wang, D.; Denton, E.; Brewer, R. "I wouldn't say offensive but...": Disability-centered perspectives on large language models. In Proceedings of the Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency, 2023, pp. 205–216. https://doi.org/10.1145/3593013.3593989.
31. Devaraj, A.; Sheffield, W.; Wallace, B.C.; Li, J.J. Evaluating factuality in text simplification. In Proceedings of the Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, 2022, pp. 7331–7345. https://doi.org/10.18653/v1/2022.acl-long.505.
32. Trienes, J.; Joseph, S.; Schlötterer, J.; Seifert, C.; Lo, K.; Xu, W.; Wallace, B.C.; Li, J.J. InfoLossQA: Characterizing and recovering information loss in text simplification. In Proceedings of the Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, 2024, pp. 4263–4294. https://doi.org/10.18653/v1/2024.acl-long.234.
33. Zheng, L.; Chiang, W.L.; Sheng, Y.; Zhuang, S.; Wu, Z.; et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In Proceedings of the Advances in Neural Information Processing Systems, 2023, Vol. 36. https://doi.org/10.48550/arXiv.2306.05685.
34. Lyon, G.R.; Shaywitz, S.E.; Shaywitz, B.A. A definition of dyslexia. *Annals of Dyslexia* **2003**, *53*, 1–14. https://doi.org/10.1007/s11881-003-0001-9.
35. Faraone, S.V.; Asherson, P.; Banaschewski, T.; Biederman, J.; Buitelaar, J.K.; et al. Attention-deficit/hyperactivity disorder. *Nature Reviews Disease Primers* **2015**, *1*, 15020. https://doi.org/10.1038/nrdp.2015.20.
36. Peng, P.; Fuchs, D. A meta-analysis of working memory deficits in children with learning difficulties. *Journal of Learning Disabilities* **2016**, *49*, 3–20. https://doi.org/10.1177/0022219414528532.
37. Rello, L.; Baeza-Yates, R. Good fonts for dyslexia. In Proceedings of the Proceedings of the 15th International ACM SIGACCESS Conference on Computers and Accessibility, 2013, pp. 1–8. https://doi.org/10.1145/2513383.2513447.
38. Ji, Z.; Lee, N.; Frieske, R.; Yu, T.; Su, D.; et al. Survey of hallucination in natural language generation. *ACM Computing Surveys* **2023**, *55*, 1–38. https://doi.org/10.1145/3571730.
39. Huang, L.; Yu, W.; Ma, W.; Zhong, W.; Feng, Z.; et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems* **2025**, *43*, 1–55. https://doi.org/10.1145/3703155.
40. Flesch, R. A new readability yardstick. *Journal of Applied Psychology* **1948**, *32*, 221–233. https://doi.org/10.1037/h0057532.
41. McNamara, D.S.; Kintsch, E.; Songer, N.B.; Kintsch, W. Are good texts always better? Interactions of text coherence, background knowledge, and levels of understanding in learning from text. *Cognition and Instruction* **1996**, *14*, 1–43. https://doi.org/10.1207/s1532690xci1401_1.
42. Kalyuga, S.; Ayres, P.; Chandler, P.; Sweller, J. The expertise reversal effect. *Educational Psychologist* **2003**, *38*, 23–31. https://doi.org/10.1207/S15326985EP3801_4.
43. Bjork, E.L.; Bjork, R.A. Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. In *Psychology and the Real World: Essays Illustrating Fundamental Contributions to Society*; Gernsbacher, M.A.; Pew, R.W.; Hough, L.M.; Pomerantz, J.R., Eds.; Worth Publishers, 2011; pp. 56–64.

44. Sweller, J.; van Merriënboer, J.J.G.; Paas, F.G.W.C. Cognitive architecture and instructional design. *Educational Psychology Review* **1998**, *10*, 251–296. https://doi.org/10.1023/A:1022193728205.
45. Baddeley, A.D.; Hitch, G. Working memory. *Psychology of Learning and Motivation* **1974**, *8*, 47–89. https://doi.org/10.1016/S0079-7421(08)60452-1.
46. LaBerge, D.; Samuels, S.J. Toward a theory of automatic information processing in reading. *Cognitive Psychology* **1974**, *6*, 293–323. https://doi.org/10.1016/0010-0285(74)90015-2.
47. Kahneman, D. *Thinking, Fast and Slow*; Farrar, Straus and Giroux, 2011.
48. Buçinca, Z.; Malaya, M.B.; Gajos, K.Z. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction* **2021**, *5*, 1–21. https://doi.org/10.1145/3449282.
49. Harp, S.F.; Mayer, R.E. How seductive details do their damage: A theory of cognitive interest in science learning. *Journal of Educational Psychology* **1998**, *90*, 414–434. https://doi.org/10.1037/0022-0663.90.3.414.
50. Gentner, D. Structure-mapping: A theoretical framework for analogy. *Cognitive Science* **1983**, *7*, 155–170. https://doi.org/10.1207/s15516709cog0702_3.
51. Spiro, R.J.; Feltovich, P.J.; Coulson, R.L.; Anderson, D.K. Multiple analogies for complex concepts: Antidotes for analogy-induced misconception in advanced knowledge acquisition. In *Similarity and Analogical Reasoning*; Vosniadou, S.; Ortony, A., Eds.; Cambridge University Press, 1989; pp. 498–531. https://doi.org/10.1017/CBO9780511529863.023.
52. Glynn, S.M. Explaining science concepts: A teaching-with-analogies model. In *The Psychology of Learning Science*; Glynn, S.M.; Yeany, R.H.; Britton, B.K., Eds.; Lawrence Erlbaum Associates, 1991; pp. 219–240.
53. Liu, Y.; Iter, D.; Xu, Y.; Wang, S.; Xu, R.; Zhu, C. G-Eval: NLG evaluation using GPT-4 with better human alignment. In Proceedings of the Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 2511–2522. https://doi.org/10.18653/v1/2023.emnlp-main.153.
54. Wang, P.; Li, L.; Chen, L.; Cai, Z.; Zhu, D.; et al. Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926* **2023**. https://doi.org/10.48550/arXiv.2305.17926.
55. Panickssery, A.; Bowman, S.R.; Feng, S. LLM evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076* **2024**. https://doi.org/10.48550/arXiv.2404.13076.
56. ACM/IEEE-CS/AAAI Joint Task Force on Computing Curricula. Computer Science Curricula 2023, 2023.
57. Cohen, J. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* **1960**, *20*, 37–46. https://doi.org/10.1177/001316446002000104.
58. Landis, J.R.; Koch, G.G. The measurement of observer agreement for categorical data. *Biometrics* **1977**, *33*, 159–174. https://doi.org/10.2307/2529310.
59. Cochran, W.G. The comparison of percentages in matched samples. *Biometrika* **1950**, *37*, 256–266. https://doi.org/10.2307/2332343.
60. Shapiro, S.S.; Wilk, M.B. An analysis of variance test for normality (complete samples). *Biometrika* **1965**, *52*, 591–611. https://doi.org/10.1093/biomet/52.3-4.591.
61. Levene, H. Robust tests for equality of variances. In *Contributions to Probability and Statistics: Essays in Honor of Harold Hotelling*; Olkin, I., Ed.; Stanford University Press, 1960; pp. 278–292.
62. McNemar, Q. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* **1947**, *12*, 153–157. https://doi.org/10.1007/BF02295996.
63. Benjamini, Y.; Hochberg, Y. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B* **1995**, *57*, 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.
64. Cohen, J. *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed.; Lawrence Erlbaum Associates, 1988.
65. Lakens, D. Calculating and reporting effect sizes to facilitate cumulative science. *Frontiers in Psychology* **2013**, *4*, 863. https://doi.org/10.3389/fpsyg.2013.00863.
66. Rose, D.H.; Meyer, A. Teaching every student in the digital age: Universal design for learning; Association for Supervision and Curriculum Development, 2002.
67. Wobbrock, J.O.; Kane, S.K.; Gajos, K.Z.; Harada, S.; Froehlich, J. Ability-based design: Concept, principles and examples. *ACM Transactions on Accessible Computing* **2011**, *3*, 1–27. https://doi.org/10.1145/1952383.1952384.
68. Treisman, A.M.; Gelade, G. A feature-integration theory of attention. *Cognitive Psychology* **1980**, *12*, 97–136. https://doi.org/10.1016/0010-0285(80)90005-5.
69. Hullman, J.; Qiao, X.; Correll, M.; Kale, A.; Kay, M. In pursuit of error: A survey of uncertainty visualization evaluation. *IEEE Transactions on Visualization and Computer Graphics* **2019**, *25*, 903–913. https://doi.org/10.1109/TVCG.2018.2864885.
70. W3C. Web Content Accessibility Guidelines (WCAG) 2.2, 2023.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.